# Supporting Information: In a Colorimetric Array of Authentic Chemical Warfare Agents, One Compound in Three Is Nearer Another Compound Than Its Own Repeat Measurement

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This Supporting Information carries the acute toxicity axis, the predictive interpolation of that axis, the decoupling test between fidelity and toxicity, the phosphorus-local reactivity analysis, the full permutation family with every multiplicity scheme, and the controls on dimensionality, subspace choice and descriptor-block weighting. These results stand on their own data and are referenced from the main text where they bear on it. They are placed here because the argument of the main text turns on the composition of the comparison panel and on whether descriptor distance predicts measured array response, and does not depend on them.

# S1 Safety axis

Acute inhalation toxicity was taken as the EPA AEGL-3 value at 10 min, the concentration above which life-threatening effects or death are predicted, parsed directly from the official compiled table;[S1] the derivation of these values for the agents is described by Watson et al.,[S2] and the concentration–time exponent on which short-exposure extrapolation of such criteria rests was established by ten Berge et al.[S3] A single fixed exposure duration was used rather than a derived concentration–time surface, and the tier value is used directly as a per-compound property rather than as an exceedance probability. Agent degradation products, which carry their own and in places distinct toxicity,[S4] are outside the compound set. Values published in ppm were converted using $c\,[\mathrm{mg\,m^{-3}}] = c\,[\mathrm{ppm}] \times M/24.45$. Compounds without a value are recorded as such, with the reason, and are never imputed.

Extraction is the step on which every safety-axis number depends, so we describe it in full. The compiled table carries 270 chemical entries that present an AEGL table of their own, besides a holding list of substances for which the agency states that values are not being developed. Reading a ten-minute AEGL-3 from one of them is not simply a matter of taking the first number on the AEGL-3 line, for four reasons that the table itself imposes. Some entries place a footnote marker in the ten-minute cell and give the value below the table, as *AEGL 3 - 10 min = ** 40,000 ppm*, because the value exceeds half the lower explosive limit. Some give values only for the shorter durations and *NR* for the longer ones, so the row carries three numbers rather than five. Some print the agency's own mg $\mathrm{m^{-3}}$ equivalent beside each value, in brackets for the agents and in parentheses elsewhere. And the text layer of the document occasionally breaks one row across two lines. We therefore parse each AEGL row into exactly five duration cells, where a cell is a number, an *NR*, a *see below*, or a bare footnote marker, and refuse to use any row that does not yield five; footnote lines are separated from table rows by grammar before parsing, and the ten-minute value is read from the text following the footnote's equals sign, so the duration labels that precede it cannot be mistaken for values. Of the 270 entries, 261 yield a ten-minute AEGL-3: 252 from a table

cell and nine from a footnote. Two of those 261, magnesium aluminium phosphide and a 60% nonafluoroisobutyl ether preparation, carry no registry number in the table and are set aside here, because without one their structure cannot be resolved unambiguously, which leaves 259 (250 from a table cell and nine from a footnote). They are not the two set aside in Section S2, which have registry numbers but no single molecular structure.

Two independent checks were applied, and the distinction between them matters. The conversion constant was checked against the agency's own bracketed mg $m^{-3}$ figures, which it reproduces to within 0.9–4.0% over eight comparisons (GB, GD, GA and GF at AEGL-2 and AEGL-3). That check validates the arithmetic and nothing else: every agent value carrying a bracketed conversion is below 1 ppm and contains no thousands separator, so a defect in numeric extraction could not appear in it. The extraction itself was therefore checked over the whole table by a second reader that shares neither the text-extraction engine nor the algorithm: it renders the document with a different tool, locates the character columns of the five duration labels in each header line, and reads the ten-minute value from the corresponding substring of the AEGL-3 line. A positional reader of this kind cannot reproduce a token-ordering fault, which is the point of using one. The two readers agree on all 250 entries for which both read a value from a table cell, on all 257 of those that resolved to a structure and on all nineteen values in the twenty-eight-compound panel. They do not agree on how many entries the table holds: the positional reader opens 269 where the primary reader opens 270, a difference of one entry that it declines to open rather than reads differently. Agreement on every compared value is therefore agreement over the 250 the two readers both reach, not over the whole table, and we state the coverage because the distinction is the one this check exists to make. Thirty-four values reach 1000 or more, where a thousands separator appears; twenty-five are read from a table cell and nine from a footnote, and 33 of the 34 are covered by the comparison against the training set. We report the coverage of this check and not only its pass rate, because a validation sample that excludes the failure mode it is meant to detect establishes nothing. An earlier version of

this check compared in one direction only, over the entries that reached the training set, and restricted the thousands-separator count to the table-cell path. Both restrictions removed the failure mode from the sample: the positional reader carried one record-boundary fault, assigning a neighbouring compound's value to ricin, and ricin does not resolve to a structure, so no one-directional comparison could have seen it. The fault is corrected, the comparison is now made in both directions, and the count covers both paths.

## S2 Predictive interpolation of the safety axis

To test whether the missing values could be filled, a random forest was trained on $\log_{10}$(AEGL-3, 10 min) using 37 RDKit descriptors, with five random seeds, five-fold cross-validation and 10 000 bootstrap resamples for interval estimates. Of the 259 entries with a ten-minute AEGL-3, 257 resolved to a structure through PubChem; the two that did not are a mixed-isomer entry and a protein toxin, neither of which has a single molecular structure. Five of the 257 are interhalogen compounds that RDKit cannot parse, and two were removed because their stored structure failed the identity check described below, leaving 250. Twenty of these are military chemical agents, identified by registry number rather than by matching on name, and all twenty were held out of training and used only for evaluation, leaving 230 for training. The held-out set is the G series including cyclosarin, VX, sulfur mustard, the three nitrogen mustards, lewisite 1, methyldichloroarsine, ethyldichloroarsine, phenyldichloroarsine, diphenylchloroarsine, adamsite, BZ, phosgene oxime, hydrogen cyanide, phosgene and chloropicrin. Matching on names rather than registry numbers is what makes such a set hard to assemble: the table lists cyclosarin as "GF Agent", which a pattern written for "Agent G" does not find, and it lists three arsenical vesicants and BZ in rows whose longer durations are *NR*, which a parser that requires five numbers per row discards entirely. External validation on held-out compounds rather than cross-validated statistics alone is the minimum standard for a predictive structure–activity model.[S5,S6] An applicability domain was defined as the

mean distance to the five nearest training compounds in standardised descriptor space, with the boundary at the 95th percentile of the training set's own such distances. This is the $k$-nearest-neighbour distance criterion whose relation to prediction accuracy was established by Sheridan et al.[S7] and which appears among the standard formulations surveyed by Netzeva et al.[S8] and Roy et al.;[S9] we have used the same construction to bound predictions in hazard-prioritisation screens.[S10] Whether such a domain can detect a compound that is close in descriptor space but outside the training set's mechanistic coverage is precisely what Section S29 tests, and is related to the open-set problem of recognising an input as unfamiliar rather than assigning it to a known class.[S11]

## S3 Decoupling test

Candidates were the AEGL compounds for which structure, electronic descriptors and a measured AEGL-3 were all available, excluding the authentic agents themselves ($n = 207$). Eight further records were set aside under four rules declared in advance, all of which ask the same question: are these electronic descriptors defined for that species at all? A record is excluded if its stored structure has more than one component, because the relative placement of the fragments is arbitrary and the dipole moment is therefore undefined (three records); if the species is open-shell, because the orbital parse reads only one spin manifold, so that its HOMO and LUMO do not carry the closed-shell meaning (three); if it is a bare atom with no bonds, because molecular descriptors carry no meaning (three); or if the HOMO–LUMO gap is below 0.5 eV, because a closed-shell single-reference description does not hold at that gap and the orbital energies themselves stop meaning anything (one). Some records fall under more than one rule. The gap rule bites on a single record: sodium cyanide returns a gap of 0.097 eV where the next smallest gap in the pool is 1.31 eV, so the 0.5 eV bar separates that one record and no other. Retaining all eight records raises the candidate count to 215 and moves the Spearman coefficient from +0.062 to +0.030, leaving every conclusion

in Section S30 unchanged. Fidelity was defined as the standardised Euclidean distance in the three-coordinate electronic space to the nearest authentic agent; safety as the measured $\log_{10}$(AEGL-3). For the quadrant analysis, the safety bar was set at the measured AEGL-3 of dichloromethane, the incumbent simulant, read from the parsed table rather than entered by hand; a replacement simulant should not be more acutely toxic than the compound it replaces. Because that bar turns out to be demanding, the count is also reported at one and two orders of magnitude below it. Association was tested with Spearman and Pearson correlation, with 10 000-resample bootstrap confidence intervals.[S12] Effect sizes are reported with intervals rather than $p$-values alone.[S13] Where a paired comparison was made, normality of the paired differences was assessed by the Shapiro–Wilk test[S14] and the Wilcoxon signed-rank test[S15] used in place of a paired $t$-test where normality was rejected; those paired tests are reported for completeness only, because, as Section  explains, they are confounded by group size and the exact permutation test supersedes them.

## S4 Conformer dependence of the shape descriptors

Nine of the coordinates in every structural space are read from a three-dimensional structure rather than from the SMILES string, so they depend on which conformer the descriptor build uses. The first version of this work took the single lowest-energy member of an ETKDGv3 ensemble of at most twenty, pruned at 0.5 Å RMSD. That choice was never stated, and it did not reproduce: rebuilding the cache from the deposited code returned eleven of fifty-two numeric columns different from the deposited values, the worst by 72 per cent of its own magnitude. The cause is that the selection is an `argmin` over near-degenerate force-field energies, so a small numerical difference between runs moves it to a different conformer.

We therefore replaced the selection with a Boltzmann-weighted mean over the whole ensemble at 298 K. Every RDKit-derived numeric column now reproduces from the code, and the descriptor build script checks them under a `-verify` flag that the analysis pipeline

requires before it will run. The scope of that gate is worth stating, because it was narrower than the pipeline's pass message once implied. It now covers the five caches the analysis reads, checking 53 columns in each, out of 53 to 62 numeric columns depending on the cache. The nine columns it does not recompute are the eight semiempirical electronic coordinates and their derivatives, which need the GFN2-xTB binary whose version was not recorded when the cache was built, together with the record of how many conformers that binary used, so the gate names them on every run rather than counting them as checked, and fails if any other column falls outside that declared list. Those eight are checked instead by the independent density-functional recomputation of Section S10. A missing cache file now stops the gate as well, because a skipped file previously left it reporting a pass with nothing compared.

The seed moves one of the eleven agents and nothing else. Repeating the embedding under six seeds and rebuilding only the nine shape coordinates, the thirty-five-compound panel returns no valid match in any of the twelve runs and the retained merged space is forty-one coordinates in all twelve. On the twenty-eight-compound panel the count is three under all six single-conformer seeds, and three under three of the six ensemble seeds against four under the other three. The agent that moves is tabun in every case. Its normalised distance runs from 1.9943 to 2.0106 across the ensemble seeds and from 2.0172 to 2.0273 across the single-conformer seeds, against a threshold of 2, so the whole spread lies within 0.03 of the threshold and straddles it. The merged-space permutation value runs from 0.0250 to 0.0279 under both treatments. Values are in the two deposited files `conformer_sensitivity_ensemble.json` and `conformer_sensitivity_single.json` under `outputs/`.

We report this at length because the conformer is a choice of the same kind as the panel, the space and the statistic, and we had not counted it. It belongs in the list of Section , and unlike the other four it does not change what the panel says about any agent except tabun. What it establishes about tabun is that this analysis does not determine whether tabun has a valid match. The embedding seed decides it, and so does the order in which the correlation

filter is handed its columns (Section S6). We report tabun as a borderline case throughout rather than counting it on either side.

## S5 The shape block as a declared sensitivity

Nine of the thirty-eight coordinates retained on the twenty-eight-compound panel, and nine of the forty-one on the primary panel, are functions of the three principal moments of inertia. Because the relations are ratios rather than linear, the correlation filter at $|r| > 0.95$ does not remove them: the largest pairwise correlation inside the block is 0.9364 on the twenty-eight-compound panel and 0.9359 on the primary panel. The one shape coordinate the filter does remove is the third principal moment itself, dropped against the second at $|r| = 0.998$ on both panels, while nine coordinates built from the same three moments are kept. They are not exact functions of one another either, since each descriptor is Boltzmann averaged over conformers independently, which leaves relative discrepancies of 1.4 to 15.3% in the three identities we checked.

We therefore kept the full block and recomputed everything with the seven derived coordinates dropped, as a declared sensitivity rather than a replacement. On the twenty-eight-compound panel the shape contribution to mean squared pairwise distance falls from 23.7 to 6.5%, and for the two agents whose nearest-match rule acetonitrile blocks it falls from 69.8 and 61.9% to 0.20 and 0.12%. The count of valid matches there falls from three to zero. On the primary panel it is zero under both encodings, and the smallest Benjamini–Hochberg value over the twelve Mantel cells rises from 0.1167 to 0.1708 with the largest rank correlation unchanged at 0.571.

## S6 Column order in the correlation filter

The descriptor spaces are built in three stages, not one, and the order is the order the code runs them. The first drops any coordinate that is constant across the panel. The second

drops one coordinate from every pair whose absolute Pearson correlation exceeds 0.95, so that no retained pair sits above that cut. The third drops any coordinate that is still an exact algebraic function of others present after that cut. On the primary panel the third stage removes nothing, because the four Abraham-parameter proxies it would otherwise catch are perfectly correlated with their determining columns and the second stage has already taken them. On the twenty-eight-compound panel it removes exactly two, `lser_S_proxy` and `is_organophosphorus`. The first of those is on the protected list and is removed anyway, because it is computed from the molar refractivity and the polar surface area and both survive; we record that the protected list does not override this stage. More than one retained set satisfies that constraint, and until this revision the set we used was chosen by the order in which the columns happened to sit in the descriptor file. The filter walked the pairs in that order and dropped the later member of each supra-threshold pair, which means the surviving coordinates were a function of a variable no analyst had selected.

The cost is measurable. Permuting the input column order forty times under a fixed seed and rebuilding everything downstream, the old filter returned thirty-four distinct retained sets on the twenty-eight-compound panel, with dimensions between 36 and 39, and a valid-match count of three in twenty-four of the forty orderings and four in the remaining sixteen. The value we had reported was the minority outcome. The thirty-five-compound panel behaved differently: thirty-six distinct retained sets and dimensions between 39 and 41, but a valid-match count of zero in all forty orderings. The headline negative result on the primary panel never depended on column order; the positive count on the reduced panel did.

One concrete case shows how the old filter reached a set it should not have. It dropped `chi0v` against molecular weight at $|r| = 0.9532$, which removed the only partner `kappa1` had above the cut, so `kappa1` survived even though $|r|(\texttt{kappa1}, \texttt{chi0v}) = 0.9816$. The retained set therefore contained a coordinate correlated at 0.98 with one the filter had just discarded as redundant.

We replaced the rule rather than fixing the order. The filter now follows the `findCorrelation`

rule of Kuhn:[S16] take the pair with the largest $|r|$, drop whichever of its two members has the larger mean $|r|$ against the other retained coordinates, recompute the correlation matrix, and repeat until no pair exceeds the cut. Three conventions make it deterministic. The input columns are sorted by name before the first pair is examined, so the caller's order cannot matter. Where exactly one member of a pair is a protected coordinate, the unprotected one is dropped, which takes precedence over the mean correlation and was a rule in the previous filter as well. Where the two mean correlations agree to within $10^{-12}$, the later name in lexicographic order is dropped, a path reached only by exactly duplicated columns.

Under the new rule the same forty permutations return a single retained set on every panel: thirty-eight coordinates and three valid matches on the twenty-eight-compound panel, forty-one coordinates and no valid match on the thirty-five-compound panel, and forty-one coordinates with five valid matches when the five literature-proposed simulants are admitted alongside the three incumbents, eight candidates in all. The maximum retained $|r|$ is 0.9470, as it was before. The counts are in `order_invariance.json` under `outputs/` and are regenerated by `order_invariance.py`, which reports both rules side by side.

The agent that moves between the two rules is tabun, and it moves because its normalised distance to dimethyl methylphosphonate crosses the threshold of 2: 1.9948 under the old filter against 2.0038 under the new one. Section S4 shows the conformer embedding seed moving the same quantity across the same threshold. Taken together, tabun's normalised distance lies within 0.03 of the threshold under every treatment we have run, and whether it counts as a valid match is decided by implementation choices rather than by the data. We therefore report three valid matches on the twenty-eight-compound panel and name tabun as a borderline case wherever the count appears, rather than presenting either three or four as settled.

## S7 Sparse indicator columns and the distance they add

Eleven of the thirty-eight retained coordinates of the merged space are binary, and they are not evenly populated. Four are carried by exactly one compound of the twenty-eight: `n_As` by lewisite 1, `n_P_CN` by tabun, `n_P_S` by VX and `n_nitro` by chloropicrin. One is carried by two compounds, three by three, two by four and one by five.

A binary column with a single one has mean $1/n$ and sample variance exactly $1/n$, so its standard deviation is $n^{-1/2}$ and standardising it places the carrier and every non-carrier exactly $\sqrt{n}$ apart. The squared Euclidean distance between the carrier and every other compound therefore gains exactly $n$, which is 28 here, from that one column, while the distance between two non-carriers gains nothing. We verified the figure against the standardised column rather than relying on the algebra; it is 28.000000.

What that addition encodes is that the compound is the only one of its kind in this panel. It is a true statement about the panel and it carries no information about proximity to any particular simulant, yet it enters the distance on the same footing as molar refractivity. The effect is concentrated on four agent-to-simulant distances. For tabun against DMMP the four single-carrier columns are 47.6% of the squared distance, essentially all of it from `n_P_CN`; for VX against dipropylene glycol monomethyl ether 20.4%; for lewisite against dichloromethane 41.8%. For sarin, soman and cyclosarin, which carry none of the four, it is zero. Counting all eleven binary columns rather than the four raises tabun's share to 72.1% and sarin's to 48.8%.

Table S1 gives the sweep. Every removal raises the count above the three of the reported encoding and lowers the nominal $p$, though the count is not monotone in how much is removed: it reaches five at one setting and returns to four at the next. The encoding we report is the one that produces the fewest matches and the largest $p$ of the seven. We state that plainly because it is the only evidence available that the encoding was not selected for its answer, and because a reader who repeats this analysis with a different descriptor set will get a different count.

Table S1: Valid matches on the twenty-eight-compound panel as sparse indicator columns are removed. The threshold for a binary column is the number of compounds carrying a one. The last row removes every coordinate with five or fewer distinct values, which also removes two counts and one protected coordinate and is the least defensible of the variants. The one-sided $p$ is the merged-space exact enumeration over all $\binom{17}{3} = 680$ designations.

| Encoding | Removed | Dim. | Matches | Tabun $d_{\mathrm{norm}}$ | $p$ |
|---|---|---|---|---|---|
| As reported | 0 | 38 | 3 | 2.0038 | 0.0250 |
| Drop single-carrier binaries | 4 | 34 | 4 | 1.4509 | 0.0250 |
| Drop binaries with $\leq 2$ | 5 | 33 | 4 | 1.4509 | 0.0206 |
| Drop binaries with $\leq 3$ | 8 | 30 | 5 | 1.6780 | 0.0162 |
| Drop binaries with $\leq 4$ | 10 | 28 | 4 | 1.4392 | 0.0132 |
| Drop binaries with $\leq 5$ | 11 | 27 | 4 | 1.2455 | 0.0132 |
| Drop cardinality $\leq 5$ | 14 | 24 | 6 | 1.1604 | 0.0074 |

The agreement test between descriptor distance and measured response does not behave this way. An earlier version of this section recomputed the twelve Mantel tests of Section under three of the encodings of Table S1 and then wrote that the result survives every choice we enumerated, which was a claim about four encodings it had not tested, including the one that moves the match count furthest. All seven are now recomputed. The smallest Benjamini–Hochberg value is 0.1167 under the baseline encoding and 0.4722, 0.3000, 0.3889, 0.2944, 0.2944 and 0.3222 under the six reduced ones, so the baseline is the most favourable of the seven and no cell reaches 0.05 under any. Removing the single-carrier indicators makes the agreement weaker, not stronger. The paper's result is in that test, and it is the one quantity here that no analyst choice we enumerated has moved.

## S8 How the designating record was searched

Section reports that we found no designating document for dichloromethane. An absence is only as good as the search behind it, so we state the search.

The NIOSH document's own reasons are worth quoting, because they are not what a detection user would assume. DMMP is called a good filtration simulant because it is relatively non-toxic, is easily detectable, and has a volatility intermediate between GB and

HD, and the properties that carry the choice are mass transfer and absorbent capacity. The National Research Council assessment of the Man-in-Simulant Test programme, which designates DMMP a second time as the fabric-testing simulant for VX, gives di-isopropyl methyl phosphonate for GB and records that the chemical structure may be quite different.[S17]

We looked for a document that names the compound as a simulant for a named agent class and states the property or measurement the choice was made against. Three sources were searched on 4 November 2026. The first is the NIOSH CBRN air-purifying respirator simulant programme, which is the document that designates dimethyl methylphosphonate;[S18] it names DMMP, carbon tetrachloride and cyclohexane as filtration simulants for GB and for HD. The latter two are not on the descriptor panel. The panel was assembled from the compounds the three source studies measured and the interferents the sensing literature names, before we traced the designation record, and we did not rebuild it afterwards; a panel that carried them would test a different question, whether the filtration designations hold on a detection axis, which no measurement here can answer. Dichloromethane is not among them. The second is the patent literature, through Google Patents and Justia, for patents naming dichloromethane as a chemical agent simulant; the training-composition patents returned name other compounds, and the one that designates dipropylene glycol monomethyl ether is.[S19] Its full text is the second limit of this search. Google Patents, Justia and the USPTO full-text service all refused retrieval from our environment, so the bibliographic record is established and the document's own wording is not; nothing in this paper rests on what that patent requires. The third is the most recent critical review of low-toxicity liquid simulants,[S20] which does not list dichloromethane as a simulant for any agent and names it only as the extraction solvent recommended by the Organisation for the Prohibition of Chemical Weapons. The fourth is the National Research Council assessment of the Man-in-Simulant Test programme,[S17] whose chapter on simulant selection names methyl salicylate, dimethyl methylphosphonate and di-isopropyl methyl phosphonate for fabric testing and does not name dichloromethane.

Table S2 lists every designation we located, with the document, the agent it names, the measurement it was made against, and the basis the document itself states. We tabulate rather than narrate because an earlier version of this section narrated it and two errors survived that would not have survived a table: it summarised the National Research Council designation of DMMP for VX as a designation for the nerve agent class, and it said elsewhere that the designating documents name no agent class for these compounds, which is false for DMMP in both documents.

One classification in the panel is contestable and we record it rather than settle it. Malathion enters the panel as an interferent, because that is how the sensing literature uses it, but the critical review cited in the Introduction assesses it as a low-toxicity simulant on the physicochemical criterion.[S20] Dichlorvos is in the same position. Neither reading changes a count. In the narrowed comparison set the only two agents either pesticide blocks are tabun, blocked by dichlorvos, and VX, blocked by malathion, and both of those agents already fail the distance threshold at 2.5320 and 3.0364, so moving the two compounds out of the comparison set leaves the three valid matches and the wider reading's zero exactly where they are.

Three limits apply and we state them rather than claim coverage we do not have. The search covers the public record reachable without clearance or a subscription, so a designation in a restricted or unindexed document would not be found. US Army Test Operations Procedure 08-2-196, *Simulant Selection for Laboratory, Chamber, and Field Testing*, is the one further document we identified as likely to carry a designation of this kind.[S21] We know of it only through the citation in TOP 08-2-188 CN1, which is also where its title comes from; a revision 08-2-196A is held by the Defense Technical Information Center under accession AD1167884. We obtained neither, so we report the procedure as unexamined rather than as absent, and we do not characterise what it contains. And the search was for a designating document, not for usage: dichloromethane is used as a simulant in the sensing literature, which is why it is an incumbent here, and that usage is not in dispute.

## S9 Lewisite purity and what depends on it

The second run reports GC–MS purities above 95% for hydrogen cyanide, cyanogen chloride, phosgene, chloropicrin, sulfur mustard and nitrogen mustard, and 67% for lewisite 1.[S22] The first and third runs report no purity values. We read the Materials section of each. The first states only that the agents were obtained from the CBRN Defense Research Institute;[S23] the third states only that the work was carried out at an Organisation for the Prohibition of Chemical Weapons designated laboratory and gives no provenance for the agents at all.[S24] Neither reports an assay, a supplier grade or a confirmatory analysis. We do not estimate the missing values.

Two things follow from the 67% and we state both. The first is the isomer. Lewisite 1 has geometric isomers and the technical material is predominantly the *trans* form. We compute it as the $E$ isomer, with input SMILES `Cl/C=C/[As](Cl)Cl`, and the descriptor code assigns that double bond `STEREOE`. Recomputing the compound as the $Z$ isomer and leaving the rest of the panel untouched moves its normalised distance in the structural space of the primary panel from 2.4769 to 2.4892, leaves CEES as its nearest candidate simulant and *para*-dichlorobenzene as its nearest non-simulant, and does not change the match verdict, which is negative under both. The comparison is of the structural space only: the electronic descriptors are produced by a semiempirical step that we did not re-run for the alternative isomer, so we do not report a merged-space figure for it. Values are in `outputs/lewisite_isomer.json`.

The second is the other 33%. The source does not say what it is, and we do not assume. Technical lewisite can carry the bis and tris homologues and arsenic trichloride, and of those the one that the sensing literature also treats as a simulant is arsenic trichloride; Section S21 adds it to the panel and reports what changes. We do not extend that to the homologues, because adding a compound the source does not place in the vial would be an assumption dressed as a control.

A 67% sample is a mixture, and lewisite enters both the descriptor panels, as one of the

eleven agents, and the second run's premise test, as one of seven compounds. Removing it changes neither. On the twenty-eight-compound panel the count stays at three valid matches with the dimension falling from 38 to 37; on the primary panel it stays at zero with the dimension falling from 41 to 39; admitting the five literature-proposed simulants, eight candidates in all, it stays at five. In the second run's premise test the mean difference moves from +0.558 to +0.529 in the ultraviolet channel and from +0.619 to +0.625 in reflectance. The sign-flip $p$ recorded at the time rose from 0.0078 to 0.0156 in both, which is that test's design floor moving with the compound count from $2^{-7}$ to $2^{-6}$ and carries no information. We withdraw the sign-flip test in Section S15, and this sensitivity does not rest on it: the statistic itself moves by 0.029 and 0.006, so removing lewisite does not change what either cell shows. The counts are in `purity_sensitivity.json` under `outputs/`.

## S10 Density-functional check on the electrophilicity axis

The electronic descriptors throughout this work are GFN2-xTB. That method is parameterised for geometries and energies rather than for virtual orbital energies, so any quantity built from its LUMO carries a question we cannot answer from inside the method. We therefore recomputed the frontier orbital energies of all twenty-eight compounds of the smaller panel at B3LYP/def2-SVPD, a basis chosen because it covers arsenic, which lewisite needs, and because its diffuse functions describe the diffuse valence density of these polar molecules. From those we formed $\mu$, $\eta$ and $\omega = \mu^2/2\eta$ by the same definitions used throughout.

Diffuse functions improve the occupied density; they do not make a virtual orbital bound. In nine of the twenty-eight compounds the B3LYP LUMO energy comes out positive, the largest $n$-hexane at +0.54 eV and the smallest acetonitrile at +0.04 eV, with dimethyl methylphosphonate at +0.076 eV and cyclosarin at +0.055 eV among them. A positive virtual orbital energy in a neutral closed-shell molecule is a discretised continuum state rather than a bound level, and its value follows the basis rather than the chemistry, so for

those nine the quantities built from the LUMO do not carry the meaning the rest of this section gives them.

The same check applies to the primary descriptors and does not clear them. Six of the twenty-eight GFN2-xTB LUMO energies are positive: dipropylene glycol monomethyl ether, tetrahydrofuran, ethanol, methanol, *n*-hexane and water. $E_{\mathrm{LUMO}}$ is one of the three coordinates of the electronic space used throughout this work, so for those six compounds one coordinate in three is an unbound virtual rather than a bound level. None of the sixteen agents is among them and one designated simulant is, so every electronic-space distance from an agent to dipropylene glycol monomethyl ether carries this. We report it rather than repair it, because repairing it would mean choosing a different electronic space after seeing what each one answers, and the counts are in `outputs/dft_electronic_space.json`. The same converged calculations also give the dipole moment, which the third paragraph of this section uses.

The comparison is not clean, and we say so rather than letting it read as one. The GFN2 values in this paper sit on GFN2-optimised geometries, which we could not reproduce in the environment where the check was run; the density-functional values sit on RDKit ETKDG geometries relaxed with MMFF94, or with UFF for lewisite, whose arsenic MMFF does not parameterise. The difference between the two sets of $\omega$ therefore contains a geometry term as well as a method term, and we do not attempt to separate them. What the check can establish is whether an ordering drawn from the GFN2 $\omega$ axis is stable, and it is not (Section S23). What it cannot establish is the size of the method term alone. All twenty-eight calculations converged; values are in the deposited `outputs/dft_omega.csv`. The dipole moments were obtained in a second pass over the same twenty-eight molecules, which reproduced both orbital energies of every compound to within $10^{-6}$ eV before the dipoles were read; that pass is deposited separately as `outputs/dft_dipole.csv` so the reproduction can be checked rather than taken on assertion.

The two sets of dipole moments agree at a Pearson $r$ of 0.9195 over the twenty-eight

compounds, with a mean absolute difference of 0.328 D. The agreement is not uniform, and the largest single disagreement falls on the compound this paper is most concerned with. DMMP is 2.523 D under GFN2 and 4.883 D under the density functional, a difference of 2.359 D, which is almost the size of the semiempirical value itself. Lewisite follows at 0.795 D against 2.108 D, consistent with the arsenic trichloride case noted below and with the force field used for that molecule, which is UFF because MMFF does not parameterise arsenic. Sarin is third at 3.640 D against 4.286 D. No other compound differs by more than 0.52 D. A reader weighing the electronic-space result of Section  should know that the coordinate most likely to move under a change of method is the one on which DMMP sits furthest from where the semiempirical method places it.

The same recomputation answers a second question, and we report the answer although it does not favour the analysis. The electronic space that carries the results of Section S16 has three coordinates: the two frontier orbital energies this check has just recomputed, and the dipole moment. The dipole follows from the same converged self-consistent field at no further cost, so there is no reason to leave it semiempirical, and we took all three coordinates from the density functional. We then repeated the matching rule of Section  and the exact permutation over all 680 designations. The control reproduces what the main text reports, one valid match of eleven and a one-sided $p$ of 0.275. Replacing the two orbital energies and leaving the dipole at GFN2 gives four valid matches and 0.128. Replacing all three gives five valid matches and 0.0368. Four of those five are sarin, soman and tabun matched to DMMP and cyclosarin matched to DPGME, pairings the designation record supports. The fifth is sulfur mustard matched to DPGME, about which the designation record says nothing either way. An earlier version of this section wrote that the patent calls DPGME a preferred G-agent simulant and rejected this pairing on that ground, and a later one wrote that the patent requires detectability by ion mobility spectrometry and assigns the compound to no agent class. We withdraw both, for the same reason: we were not able to retrieve the text of that patent, so nothing about its internal requirements is established here and neither its

agent class nor its detection instrument enters any argument (Section ). The single match under GFN2 was hydrogen cyanide to dichloromethane.

Three things have to be said about that. The first is that the sign of the conclusion in this one space depends on the electronic-structure method. Under GFN2 the electronic space was where the incumbents did worst and nothing in it came near significance; under the density functional the same space, the same matching rule and the same exact permutation return 0.0368. We do not promote that value to a positive finding. It is a single uncorrected statistic in one space, reached by changing a method after seeing the first answer, and the geometry confound of the preceding paragraph applies to it in full. We report it because the alternative is to leave standing a negative claim that our own check does not support. The second is that the merged space does not move the same way. Rebuilt from the same three density-functional coordinates it retains the thirty-eight coordinates of the main analysis and returns three valid matches and a one-sided $p$ of 0.0250, against three and 0.0250 for the GFN2 control, so the value the main text carries is not an artefact of the semiempirical method. The third is that the analyst freedoms this paper is about therefore include one we did not enumerate when we set the comparison up, and it is the freedom that moved the electronic space furthest of any we examined. Values are in the deposited `outputs/dft_electronic_space.json`.

We did not extend the check to the condensed Fukui index. The only population scheme available to us at this level is a Mulliken partition, and on a diffuse basis that can return a negative condensed index for electrophilic attack, which is an artefact of the partition rather than a chemical statement. We therefore have no independent check of this quantity. The $f^{+}$(P) values in Table S10 are therefore GFN2 values with no independent check, and the ratio between DMMP and VX that the main text quotes should be read within that limit.

## S11 The seven analyst freedoms in full

The main text reports that seven analyst choices move the count of valid matches and keeps only the summary. The three that are not the panel, the filter order, the threshold or the comparison set are given here as they stood in the body, followed by the comparison set, which was added in this revision.

The comparison set is the one the second condition is tested against. The Experimental Section of the main article names every compound outside the designated set, and the agents are outside it, so the literal reading places the other agents in the comparison set and asks whether an agent's nearest designated simulant is nearer to it than every other agent is. The deposited audit now reports three readings. The wide reading, every compound that is neither an agent nor a designated simulant, is what the primary analysis uses and gives three valid matches on the twenty-eight-compound panel, none on the primary panel and five on the panel that adds the five literature candidates. The narrow reading, interferents only, agrees with it on the twenty-eight-compound panel, where the two sets coincide, and gives three on the primary panel. The literal reading gives zero, zero and two. Under the legacy filter the twenty-eight-compound count is four, four and one. The literal reading is the largest single move in this list, and the direction is against the designations in every case.

One analyst freedom we did not enumerate at the outset moved the answer further than any we did. Recomputing the three electronic coordinates at B3LYP/def2-SVPD instead of GFN2-xTB moves that space from one valid match of eleven and a one-sided $p$ of 0.275 to five matches and 0.0368, and only one of the five is a pairing the designation record makes. We do not promote it: one uncorrected statistic in one space, reached by changing a method after seeing the first answer, on different geometries. What it establishes is that the space in which the incumbents did worst is the space the choice of electronic-structure method decides. The merged space keeps three valid matches and a one-sided $p$ of 0.0250 under both methods while its individual distances move (Section S10).

So does the way the descriptors are encoded, and that freedom is the largest of the six.

Four of the thirty-eight retained coordinates on the twenty-eight-compound panel are binary indicators carried by exactly one compound, and standardising such a column adds exactly $n$, here 28, to the squared distance between its carrier and every other compound. For tabun that addition is 47.6% of the squared distance to DMMP. Dropping those four moves tabun from 2.0038 to 1.4509 and the count from three valid matches to four; dropping every indicator carried by three compounds or fewer gives five, and every coordinate with five or fewer distinct values gives six, with the merged-space one-sided $p$ falling from 0.0250 to 0.0074. We kept the encoding that produces the smallest count and report the sweep, because that is the only evidence that we did not select the encoding for its answer. Section S7 gives the column-by-column accounting.

A fifth choice is the shape block: nine retained coordinates on each panel are functions of the three principal moments of inertia, which the correlation filter does not remove because the relations are ratios rather than linear. Dropping the seven derived ones, as a declared sensitivity rather than a replacement, takes the twenty-eight-compound count from three to zero, and the primary panel is zero either way (Section S5).

The sixth freedom is the match threshold. The criterion of Section  has two parts, a normalised distance of 2 or less and a distance smaller than that to every compound outside the designated set. Removing the first part and keeping the second takes the twenty-eight-compound count from three valid matches to five, while the primary panel stays at zero; the null expectation of the number of agents with something outside the designated set nearer than any incumbent is 9.06 of eleven on the twenty-eight-compound panel and 9.63 of eleven on the primary one, so what the count is sensitive to is the threshold and not the ordering the second part encodes. The values are in `outputs/stat_supplement.json` under `threshold_free`.

## S12 Multiplicity and the accounting schemes

The designation tests of the main text are one-sided, because the hypothesis is directional. The two-sided values matter on the twenty-eight-compound panel, where the convention decides the answer: the structural space moves from 0.0279 one-sided to 0.0559 at twice the smaller tail and 0.0441 scored as distance from the null mean, and the merged space from 0.0250 to 0.0500 and 0.0279. Under the first of those conventions neither space reaches 0.05.

Multiplicity is corrected within each panel configuration. The six distance statistics applied to the three descriptor spaces form one family of eighteen tests on the same descriptor table, and we report the whole family rather than a selection from it, so the Benjamini-Hochberg procedure[S25] is applied across those eighteen and the smallest adjusted value is given in Table S4. An earlier version of this work reported one nominal value per space and declined to correct, on the ground that the spaces were separate analyses. That ground fails once the statistic is also chosen by the analyst, and we treat the earlier position as an error rather than a convention.

Defining that family is itself a choice, and a paper arguing that such choices decide the answer cannot leave its own unexamined. We therefore report four schemes rather than one. The eighteen tests are corrected together; the six statistics are corrected within each space separately, treating the spaces as three analyses; a single statistic, the mean difference, is treated as though it had been specified in advance, leaving three tests; and, because the three spaces are nested and the six statistics are functions of one distance matrix, the dependence is handled exactly rather than approximated. All eighteen tests are computed on the same enumeration of designations, so the joint null distribution is available: for each designation we convert every test statistic to its one-sided $p$ within its own null, take the smallest of the eighteen, and read the observed smallest $p$ against that distribution. This is the single-step min-$P$ procedure of Westfall and Young,[S26] and because the permutations are shared it requires no assumption about the correlation structure. We also report the approximation it replaces, the effective number of independent tests estimated from the eigenvalues of the

correlation matrix of the eighteen nulls by the method of Li and Ji[S27] used as a Bonferroni divisor, so that the two can be compared. The five answers are given in Table S5 and they do not agree.

## S13 Dimensionality and subspace controls

The structural and electronic spaces differ in dimension by more than an order of magnitude, and the concentration of Euclidean distances depends on dimension, so a difference between the two spaces is not by itself a difference between chemistries. We bounded the effect in two ways, both reported in Section S18. The standardised structural space was truncated to its first three principal components and the permutation test rerun on that three-dimensional space. Separately, 2000 random three-descriptor subsets of the structural space were drawn without replacement (seed `0xC0DE`) and the full exact permutation test run on each, giving the distribution of $p$ that an arbitrary three-dimensional slice of these descriptors produces. The observed structural value is interpretable only against that distribution.

The random-subspace control of Section S13 draws three descriptors from a correlated pool, so three descriptors do not carry three degrees of freedom and the resulting rate of nominal significance may be inflated. We repeated it on ZCA-whitened coordinates, obtained by standardising the structural block, taking the eigendecomposition of its covariance, and applying $V\Lambda^{-1/2}V^{\top}$ over the non-null eigenvalues, so that the coordinates drawn from are uncorrelated by construction. Both the raw and the whitened rates are reported. That draw count carries a Monte Carlo standard error near one percentage point on a proportion and resolves a percentile to 0.05.

## S14 How the sixteen agents were classified under the Convention

The main text states that twelve of the sixteen measured agents are Chemical Weapons Convention Schedule 1 compounds and four are Schedule 3 industrial chemicals. The assignment is read from the Annex on Chemicals as the Organisation for the Prohibition of Chemical Weapons publishes it.[S28] Sarin and soman fall under 1A(1), tabun under 1A(2), VX under 1A(3), sulfur mustard under 1A(4), lewisite under 1A(5) and nitrogen mustard under 1A(6); cyclosarin is not named individually and falls under 1A(1), whose $R_2$ position admits a cycloalkyl group. Phosgene, cyanogen chloride, hydrogen cyanide and chloropicrin are 3A(1) to 3A(4).

The four A-series compounds fall under Schedule 1 entries added by the 2019 amendment, which entered into force on 7 June 2020.[S29] We assign no particular entry to any of them, for two reasons. The amendment added entries 1A(13) to 1A(16), and those four do not stand in one-to-one correspondence with these four codes: 1A(15) names a single compound and 1A(16) covers a family of carbamates rather than organophosphorus compounds. The source study moreover reports the compounds as the codes A-230, A-232, A-234 and A-242 and gives their structures as representative rather than assigned (Section ), so the entry that applies to each code is not fixed by anything available to this reanalysis. Their Schedule 1 membership is stated at the level of the amendment and not of a named entry.

## S15 Measured colorimetric array response

Measured response was taken from a 29-component colorimetric array reported in earlier work from this laboratory.[S22–S24] The array is listed compound by compound in Table S3. Six of the twenty-nine are not chromogenic dyes: three are fluorophores, one is a thiol reducing agent and two are carbonyl derivatising reagents, so part of the pattern is covalent

derivatisation or emission rather than a solvatochromic shift. The three fluorophores are quinine, anthracene and pyrene, and they are the components whose response is most asymmetric between the two channels: quinine reaches a maximum $\Delta E_{00}$ of 90.2 in the ultraviolet channel against 3.6 in reflectance, a ratio of 79.6, the largest of the twenty-nine, and anthracene and pyrene follow at 24.7 and 14.7. The ratio is computed for every component in `outputs/dye_panel_audit.json`, so where the line falls is visible rather than asserted. The stock solvent of one component, Quinoline Yellow, differs between the source studies, dimethyl sulfoxide in the first two runs and water in the third; we report that as published and do not resolve it. Colour difference $\Delta E$ is the CIEDE2000 difference against the zero-concentration image of the same dye in the same channel. The source studies do not state the working colour space, the reference white or the transfer function used to reach CIELAB from the camera output, and we did not have the plate images, so that conversion is an input to this reanalysis rather than part of it. Every comparison made here is within one run and one channel, where whatever convention was used is common to the vectors being compared and cancels from the cosine; no comparison in this paper puts a $\Delta E_{00}$ from one run beside one from another. The values are given given per dye for three compound sets at solution concentrations of 10, 50, 100 and 500 $\mu$M delivered in acetonitrile, in two imaging channels: ultraviolet-excited emission and reflectance. An earlier version of this section called these nominal vapour concentrations in arbitrary units, which was wrong on both counts; the source reports read as solution concentrations in micromolar, and the correction is recorded here because the concentration axis carries the trend reported in Section S24. The sets are DMMP with GA, GB, GD, GF and VX; AC, CG, CK, HD, HN and L with PS; and four A-series agents. The first two sets supply the twelve compounds that are also in the descriptor panel: six of the six in the first set and six of the seven in the second, the exception being nitrogen mustard HN, which the descriptor panel does not carry. The third set is excluded entirely and the criterion is arithmetic rather than editorial: none of its four compounds appears in the descriptor panel, so no pair of a measured pattern and a

descriptor distance can be formed from it. It contributes to no analysis in this paper. The exclusion is not a judgement we had to make on chemical grounds, because the source does not supply the identities that such a judgement would need: it carries the four as the codes A-230, A-232, A-234 and A-242, labels its structure figure representative, and does not map any code to a named compound.[S24] We therefore do not assign structures to those codes anywhere in this work.

The three sets are different compounds measured on different plates, not replicates, so no $\Delta E$ value is compared across sets; every analysis is run within a set and the results are reported separately or combined only at the level of correlation coefficients. The two channels are likewise not replicates but physically different measurements, and the same compound's patterns in the two channels agree only to a mean cosine of 0.353 over the seventeen compound-and-plate instances, computed on the centred patterns every test here uses, and 0.584 on the unit-scaled patterns before centring. They are therefore treated as independent modalities, and agreement between them is used as evidence of reproducibility rather than averaged away.

Each compound's 29-dimensional response at each concentration was scaled to unit length, so that pattern rather than magnitude is compared. The compound-axis mean was then subtracted at each concentration to remove the common non-negative response component, and each centred vector was scaled to unit length again. Those three steps, in that order, are what the code performs, and the second scaling is load-bearing for the argument two paragraphs below. The resulting vectors were averaged over concentrations and renormalised to give one pattern per compound per channel. Similarity is the cosine between patterns and dissimilarity is one minus that. Confidence intervals are bootstrapped over concentrations with 10 000 resamples, which means they describe variation across the four concentrations of one plate set and not agreement between independent preparations.

Two properties of this processing constrain what the patterns can be used to show, and we state both because the first version of this analysis did not.

Subtracting the compound-axis mean at each concentration makes the resulting vectors sum to zero across compounds. An earlier version of this section drew from that an identity fixing the mean cosine between distinct compounds at $-1/(n-1)$ for $n$ compounds in a set, and compared the observed means against it as against a floor. Two things are wrong with that, and we state both because the comparison appeared in the main text.

The first is that the processing does not stop at the subtraction. Each centred vector is renormalised to unit length before any cosine is taken, and renormalisation does not preserve $\sum_i u_i = 0$ unless the norms happen to be equal. Measured on the vectors this work actually uses, $\|\sum_i u_i\|/\sqrt{n}$ is $7.2 \times 10^{-16}$ immediately after the subtraction and between 0.0445 and 0.7430 after the renormalisation. The premise of the identity is not merely perturbed, it is gone.

The second is that $-1/(n-1)$ is not a floor even for vectors that do sum to zero. From $\|\sum_i u_i\|^2 = 0$ one has $\sum_{i<j}\langle u_i, u_j\rangle = -\frac{1}{2}\sum_i \|u_i\|^2$, while $\sum_{i<j}\|u_i\|\|u_j\| \le \frac{n-1}{2}\sum_i \|u_i\|^2$ by Cauchy–Schwarz. Dividing the first by the second gives a norm-weighted mean cosine of at most $-1/(n-1)$, with equality only when every norm is equal. The quantity is bounded above by that value, so a margin above it was never the thing to read.

We therefore do not compare the observed means against $-1/(n-1)$ anywhere in this work. The value is reported in `outputs/premise_null.json` for comparison only. Section S24 compares the observed means against a permutation null instead: compound labels are permuted within each concentration, which leaves the set of centred vectors in a concentration unchanged, and therefore their mean unchanged, while changing which compound each one belongs to. That is the null the premise claim needs, because the claim is that one compound's response at one concentration is related to the same compound's response at another.

Pairwise distances are not independent observations. A set of $n$ compounds gives $\binom{n}{2}$ pairs that share compounds, so correlating measured dissimilarity against descriptor distance over pairs and testing it as if the pairs were independent inflates significance; the unit of analysis

is the compound. We therefore test the association by Mantel permutation,[S30] permuting compound labels on the descriptor distance matrix over all $6! = 720$ relabellings of the six compounds each cell carries, and comparing the observed Spearman coefficient against $|\rho|$ of the permuted matrices. The enumeration over this family of twelve cells is exhaustive, so these twelve tests are exact and carry no sampling seed. The analyses in this document that are sampled rather than enumerated, the Mantel power curve, the null match expectation, the premise null in its four larger cells, the collapse sensitivity and the reversal audit, are declared with their seeds and their numbers of draws where they appear. Both the Mantel value and the value that treating pairs as independent would have given are reported, so the size of the error is visible.

## S16 The full permutation family

Table S4 gives every exact permutation test we can run across four panel and designation configurations, three spaces and six statistics, in full rather than selected from, because the pattern across the cells is the finding. Main-text Section  states what holds across it.

Two further values belong with Table S5. Weighting the 35 structural and 3 electronic descriptors as two equal blocks, instead of the per-descriptor standardisation used throughout, lowers the primary panel's smallest nominal value from 0.192 to 0.061, which still survives no correction and leaves the primary panel's valid-match count unchanged at zero. On the twenty-eight-compound panel the two weightings do not agree, four matches per descriptor against three per block, which is one more way the smaller panel's count turns on a choice nobody declares (Section S20). The Li-Ji approximation that the Westfall-Young procedure replaces estimates 5.00 effective tests of eighteen and gives 0.125, conservative here by a factor near 1.7. None of the first three configurations of Table S4 reaches 0.05 under any of the four schemes that correct what was actually run.

The sixth row was added after a reading of an earlier draft pointed out that the first

five all correct within one panel configuration, while the paper's own argument is that the configuration is itself a choice. We therefore pooled the four configurations into one family of 72 tests and corrected across all of them. The smallest adjusted value on the twenty-eight-compound panel rises from 0.163 to 0.236 and the two thirty-five-compound configurations rise to 0.354 and 0.236, while the relabelled configuration stays significant at 0.003. Nothing in the paper's conclusions changes, and the scheme that most nearly reaches significance is still the one that fixes the statistic in advance. We report it because leaving it out would have exempted our own accounting from the standard the paper asks of others. Values are in the deposited `multiplicity_across_panels.json` under `outputs/`.

## S17 The volatility axis

The opening of this paper names volatility as the first property a simulant is chosen to reproduce, and the descriptor panel does not contain it. That is a gap in the analysis and not a point we can argue away, because a reader is entitled to ask whether the null we report is a property of simulant selection or a property of the particular descriptors we happened to compute. This section closes it with two tests fixed before the numbers were seen. Both are reported outside the family of seventy-two, and we say so here rather than fold them in, because folding them in would recompute every adjusted value in the paper in exchange for a question that one test each can answer.

Vapour pressures were taken from measurements, not predictions. The adoption rule was fixed before any value was chosen: a vapour-pressure measurement or compilation at 25 °C first, a government compendium at 25 °C where no such value exists, and the nearest available temperature otherwise, recorded with its temperature. On the twenty-eight-compound panel all twenty-eight compounds have a measured value, twenty-four of them at tier one, one at tier two and two at tier three. The two tier-three entries are dipropylene glycol monomethyl ether at 26 °C and cyclosarin at 20 °C, for which no 25 °C measurement is in the public record we

searched. One further entry, ethanol, is marked TX rather than tier one because the source that supplies its 25 °C value labels it as extrapolated rather than measured; it is an interferent and enters no pairing reported here, and we flag it rather than adopt a measured value at a different temperature. We did not convert between temperatures, because conversion needs Antoine coefficients and fitting them would replace a measurement with an estimate. On the thirty-five-compound panel two compounds have no measured vapour pressure at all, diethyl chlorophosphate (CAS 814-49-3) and diethyl cyanophosphonate (CAS 2942-58-7), so the volatility tests run on the twenty-eight-compound panel only. Table S6 lists the adopted values with their tier and source; every candidate value returned, including the ones not adopted, is in the deposited `volatility_candidates.json`.

The space is one axis, $\log_{10}(P/\mathrm{mmHg})$ standardised to unit variance, and distance is the absolute difference. One axis removes three of the four analyst freedoms this paper measures elsewhere, because the descriptor encoding, the correlation filter and the conformer seed have nothing to act on. It introduces one of its own, and we report it rather than claim the axis is choice-free. Which measurement is adopted where a compound has more than one is a choice, and it moves the result: taking the *cis* isomer value for lewisite, 1.562 mmHg at 25 °C from the same source that gives the mixture value, moves the pre-specified statistic from $p = 0.0206$ at rank fourteen to $p = 0.0103$ at rank seven. Dimethyl methylphosphonate carries a competing 25 °C value of 1.2 mmHg against the adopted 0.962, which is 0.097 decades away, nine times the gap on which the single valid match rests. The adoption rule above was fixed before the values were seen, which is why we state it, but it is a choice and not an absence of one. The panel spans 6.209 decades, from VX at the low end to phosgene at the high end.

The first test asks whether the three incumbent simulants sit closer to the agents on this axis than a trio drawn from the seventeen non-agents would. Under exact enumeration of all 680 designations, the pre-specified statistic gives a one-sided $p$ of 0.0206, ranking the incumbent trio fourteenth of 680. That is not a null, and we report it as the one place in

this paper where the incumbent choices beat chance. Two things bound what it means. The margin itself is 0.0015 in standardised units, which is not a separation anyone could use. And the count of valid matches, the criterion this paper argues a designation has to meet, returns one match with a $p$ of 0.4647, ranking 316th of 680 and therefore indistinguishable from chance.

That single match is dipropylene glycol monomethyl ether to soman, and it rests on a gap of 0.0107 decades between a measurement at 26 °C and a measurement at 25 °C. The gap is smaller than the temperature mismatch that produced it. Putting both compounds at a common temperature can only widen it, because vapour pressure increases with temperature and the 26 °C figure of 0.4 mmHg therefore overstates the 25 °C value; the direction is fixed thermodynamically and needs no fitted correlation. The match fails the ratio condition once the gap exceeds 0.1383 decades, which corresponds to a 25 °C vapour pressure below 0.2982 mmHg, and fails the closer-than-every-interferent condition beyond 0.1654 decades, or 0.2802 mmHg. We do not have a measured 25 °C value for that compound, so we report the match as conditional on a gap that its own source data cannot resolve.

Table S7 sets out which compound each agent is nearest to. The axis does not reproduce the pairings the usage literature implies. Dichloromethane sits 0.514 decades from phosgene but 0.232 from hydrogen cyanide. Dipropylene glycol monomethyl ether sits 0.561 decades from sulfur mustard and 0.011 from soman. Dimethyl methylphosphonate, the compound the field treats as the G-agent surrogate, sits 0.473 decades from sarin while *para*-dichlorobenzene, an interferent, sits 0.216. For three of the eleven agents the nearest compound in the pool is an interferent rather than a simulant. VX has nothing within 2.659 decades. We state these as distances and not as departures from designated pairings, because the designating documents do not pair these compounds with agents on anything a volatility distance can be read against. Dimethyl methylphosphonate is the one exception and it points two ways at once: the NIOSH programme names it for GB and for HD against a filtration measurement, and the National Research Council assessment names it

for VX against a fabric test. Dipropylene glycol monomethyl ether appears in a training-composition patent whose text we could not retrieve, so it pairs with no agent here either, and for dichloromethane we found no designating document at all. A distance to sarin is therefore not a departure from a designated pairing for two of the three, and for the third it would have to be read against two different agents at once.

The second test asks whether volatility distance tracks response distance. Exact Mantel permutation in each run-channel cell with at least four mapped agents gives four evaluable cells, and the rank correlation is positive in all four: +0.043 and +0.254 in the ultraviolet channel, +0.389 and +0.546 in the reflectance channel. A single Fisher combination over all four cells was fixed in advance as the statistic for this test. It returns $p = 0.0809$, and we withdraw it, but the order of those two facts has to be stated plainly because it bears on everything below: the prespecified statistic did not clear 0.05, and the family we report instead was chosen afterwards. The ground for withdrawing it is methodological and not the value. The two optical channels of one plate are read from the same wells and are not independent, with a mean inter-channel cosine of 0.353 across the array on the centred patterns and 0.584 before centring, so the four $p$ values do not satisfy the assumption that combination requires. Runs do. We therefore combine across runs within each channel and correct across the two channels with Benjamini-Hochberg. The reflectance channel gives $p = 0.0215$ and $q = 0.0430$; the ultraviolet channel gives $p = 0.6389$ and $q = 0.6389$. Read as one family of four cells instead, the smallest adjusted value is 0.1389, and corrected jointly with the twelve descriptor cells it is 0.1556. The $q = 0.0430$ is therefore a property of one post-hoc family out of three we can define on the same four numbers, and we report all three rather than the one that clears. The withdrawn pooled statistic is recorded in `outputs/volatility_space.json`.

Taken together the two tests say something narrower and more useful than the descriptor result alone. The axis practitioners name first does carry directional signal, significant in the reflectance channel, and the incumbent trio does beat chance on margin statistics, so the

paper's null is not an artefact of leaving volatility out. What the axis does not produce is a defensible valid match, and the one match it does produce rests on a gap its own source data cannot resolve. The conclusion of the paper is unchanged and its basis is wider.

## S18 Dimensionality and the value of the nominal result

The structural space has 35 coordinates and the electronic space has 3. Euclidean distance behaves differently in the two, so a difference between them could be geometric rather than chemical. We tested that, and it is not. We also tested what the nominal structural value is worth against the descriptors it was drawn from, and there the answer is less comfortable.

Truncating the standardised structural space to its first three principal components does not weaken the result; it strengthens it, from $p = 0.028$ to $p = 0.0029$ on the twenty-eight-compound panel. At matched dimension the two spaces are still 94-fold apart in $p$ (0.0029 against 0.275). Whatever separates the structural from the electronic result, it is not that one space has more coordinates than the other, and we say so because the opposite is the natural suspicion and because an earlier draft of this paper entertained it.

The second control is the informative one. We drew 2000 random three-descriptor subsets of the structural space and ran the full exact permutation test on each, taking the mean-difference statistic throughout, so that the draws and the observed value are the same quantity. On the twenty-eight-compound panel the median $p$ is 0.149, the 5th to 95th percentile range is 0.006 to 0.710, and 28.95% of the subspaces reach $p < 0.05$ (Monte Carlo standard error 1.08 percentage points). The observed 35-dimensional value of 0.028 sits at the 22.4th percentile of that distribution, and the electronic value of 0.275 at the 68.8th. The three-principal-component value of 0.0029 sits at the 1.6th percentile, which is what the leading components of a space should do and is not independent evidence.

## S19 The subspace share needs a null, and we did not give it one

An earlier version of this paper reported the share alongside the percentile and read both as showing that the twenty-eight-compound structural result is an ordinary draw. The percentile supports that reading. The share does not, and the reason is that we never asked what the share would be if the designation carried nothing.

In a fixed three-descriptor subspace the permutation $p$ of a designated set is its rank among all 680 ways of designating three of the seventeen non-agents. If the designation were arbitrary that rank is uniform, so the expected share of subspaces reaching $p < 0.05$ is 0.05. The observed 27.4% is 5.5 times that.

One enumeration of a subspace yields the statistic for all 680 designations at once, so the null costs nothing beyond the run already made. Over 500 subspaces drawn the same way, the share below 0.05 averages 4.5% across the 680 designations, with a median of 2.4%, a 95th percentile of 17.8% and a maximum of 44.6%. The incumbent designation's 27.4% ranks seventh of 680, $p = 0.0103$. The subspaces are drawn from the thirty-five structural coordinates the correlation filter of Section S6 retains. An earlier version of this section reported this family from a thirty-seven-coordinate space that predated that filter, and the share there was 36.0% at rank five. Values are in the deposited `subspace_null.json` under `outputs/`.

We therefore withdraw the use of this share as evidence against the twenty-eight-compound structural cell. The structural block does separate the incumbent triple from an arbitrary one. Two things bound what that means. The observed mean difference is +0.155, so the designated simulants are on average further from the agents than the nearest compound outside the designated set; the test says they are less far than an arbitrary triple, not that they are near. And the arbitrary triple is drawn from fourteen interferents and solvents that entered this panel because the sensing literature treats them as the things a detector must

not confuse with an agent. Being less unlike the agents than that reference class is a low bar for one organophosphorus compound to clear. The result that does not depend on the reference class is the thirty-five-compound one, where compounds the literature itself calls simulants are in the pool and nothing reaches significance at all.

That rate is partly an artifact of the descriptors themselves, and we report the correction because it is larger than we expected. Three descriptors drawn from a correlated pool do not carry three degrees of freedom. Repeating the control on ZCA-whitened coordinates (Section S13), where the coordinates are uncorrelated by construction, the rate falls from 28.95% to 17.25% and the median $p$ rises from 0.149 to 0.332. The correlation structure of the structural block therefore inflates the apparent ease of reaching nominal significance by more than a factor of two. The comparison that the argument rests on is the percentile rather than the rate, and that is unaffected: the observed value is an ordinary draw from the subspaces that were available to be chosen.

So the structural and electronic spaces do differ, and the difference is chemical rather than geometric; what the data do not support is that the difference says anything about the compounds that were designated. A value at the 22.4th percentile of the subspaces one could have chosen, in a family of eighteen tests whose smallest adjusted value is 0.163, on the smaller of two defensible panels, is not an argument. Picking a space, a panel and a statistic fixes the answer, and the practice this paper examines has never written down which to pick.

## S20 Weighting the two descriptor blocks equally

The merged space puts 35 structural coordinates and 3 electronic ones into one Euclidean distance. Standardising each descriptor equalises the variance per coordinate, not per block, so the expected contribution is 38 against 3 and the merged distance is in practice a structural distance. The main text says so; this section quantifies it, because a reader is entitled to ask whether the merged result is an artifact of that imbalance. A colleague who read a draft

asked exactly this.

We repeated the whole analysis on a distance that equalises the expected contribution of the two blocks,

$$d^2_{\mathrm{bal}} = \frac{1}{n_s}\sum_{i=1}^{n_s}(\Delta z_{s,i})^2 + \frac{1}{n_e}\sum_{j=1}^{n_e}(\Delta z_{e,j})^2, \tag{1}$$

with $n_s = 35$ and $n_e = 3$, leaving the structural and electronic spaces untouched since each is a single block. The same exact enumeration was used, over the same three configurations, with the same six statistics.

The valid-match counts do not move under either weighting, on any panel construction: zero of eleven on the primary panel with the incumbent three, five with all eight candidates admitted, and three on the twenty-eight-compound panel. The paper's headline is therefore invariant to the weighting, and so is the pre-specified-statistic counterfactual of Table S5, which stays at 0.042 on the twenty-eight-compound panel: the merged nominal value there falls from 0.025 to 0.016, but the value that binds after correction is the structural one, which the reweighting does not touch. No configuration has a Benjamini–Hochberg adjusted value below 0.05 under either weighting.

Every nominal $p$ falls, however, and that is why the main text states the primary-panel result as a conclusion about significance rather than as a numerical floor. The smallest value of the eighteen-test family there is 0.192 with equal weight per descriptor and 0.061 with equal weight per block, so a floor quoted from either weighting would not hold under the other. The direction is away from the designations: equalising the blocks makes the incumbent simulants look better on the nominal scale while leaving them with no agent matched. Block weighting is therefore a fourth choice of the same kind as the panel, the space and the statistic, and it was not declared in advance either. We report it because we did not pre-register it, and because a reader who did would have obtained numbers we would not otherwise have published.

## S21 A chemically grounded lewisite simulant

Of the eleven agents, lewisite is the one whose incumbent simulant is chemically least like it: dichloromethane shares neither the central element nor the halogen count. A reader may reasonably ask whether the lewisite result reports a failure of the designation or only a poor choice of comparator on our part. We therefore repeated the panel analysis with arsenic trichloride added. The compound is used as a lewisite simulant in the sensing literature.[S31] We found no designating document for it, so it enters as a literature simulant, on the same footing as the other five, and not as an incumbent. Descriptors were generated by the same procedure as the rest of the panel, without modification. The optimised geometry is pyramidal, with Cl–As–Cl angles of 98.8° against a reported 98.4°.

The incumbent-three count is unchanged at zero of eleven. This is not a coincidence of these data. A compound that is not a designated simulant can enter the matching rule only through its second clause, as one more non-simulant that a designated simulant must beat; adding one can make a match harder and never easier, and the count is already at its floor.

The single match the addition creates is lewisite's. Under the literature set without arsenic trichloride the nearest literature simulant to lewisite is 2-chloroethyl ethyl sulfide, a sulfur mustard simulant, at 2.35 and outside the threshold: the five contained no candidate built on the right element. Arsenic trichloride enters at 1.25 and matches. We draw no recommendation from this, for the reason given in the main text. The point is narrower. For one agent the compound the descriptor space selects and the compound that carries the designation are different compounds, and the designation is the one for which we found no document.

**Limitation.** The semiempirical method returns a dipole moment of 0.06 D for arsenic trichloride against a reported value near 1.6 D. The method is applied identically to every compound in the panel, so relative positions remain comparable, but absolute electronic descriptors for heavy main-group compounds should not be read on their own.

## S22 What the test could have detected

A null result invites the question whether the design could have seen an effect at all, and the exact enumeration answers it without further assumption. We moved the incumbent simulants closer to the agents by $\delta$, measured in typical nearest-neighbour separations of the same space, left the rest of the panel untouched, and read the shifted statistic against the null built from the unmodified distances. The reading is conditional: on this panel, with only the designated three made more faithful, the test returns $p < 0.05$ at $\delta = 0.23$ (Figure S1). At $\delta = 0$ it returns 0.2979 on the primary panel and 0.0250 on the twenty-eight-compound panel, reproducing Table S4.

The comparison that matters is between that value and the deficit the incumbent compounds actually carry. Across the eleven agents the normalised difference between the nearest incumbent simulant and the nearest other compound has a median of 0.4926. Against the uncorrected threshold of 0.23 that is more than twice the detectable value, but the threshold this paper is entitled to use is the corrected one. Requiring the shifted statistic to clear the Benjamini–Hochberg level of the eighteen-test family moves the threshold to $\delta = 0.43$, and the median deficit is then 1.15 times what the design can resolve rather than twice it. On the twenty-eight-compound panel the corrected threshold is $\delta = 0.18$ against a median gap of 0.074, a ratio of 0.41, so on that panel the design cannot resolve the deficit the compounds carry and we do not read its null as a statement about the data.

Sarin is the one agent whose deficit, 0.125, falls below even the uncorrected threshold. It is not the agent at which the descriptor resampling of Section  is weakest: DMMP recovers sarin as its nearest agent in 89.8% of 1000 bootstrap replicates, the highest of the three incumbents, where DPGME recovers sulfur mustard in 41.0% and dichloromethane recovers cyanogen chloride in 47.5%. The sarin case is therefore a resolution limit of the design and not an unstable estimate. Read with the corrected threshold, the primary-panel null is a statement about the data for the agents whose deficit exceeds 0.43 and a statement about sensitivity for the rest, and we mark that boundary rather than claiming the whole panel

for either.

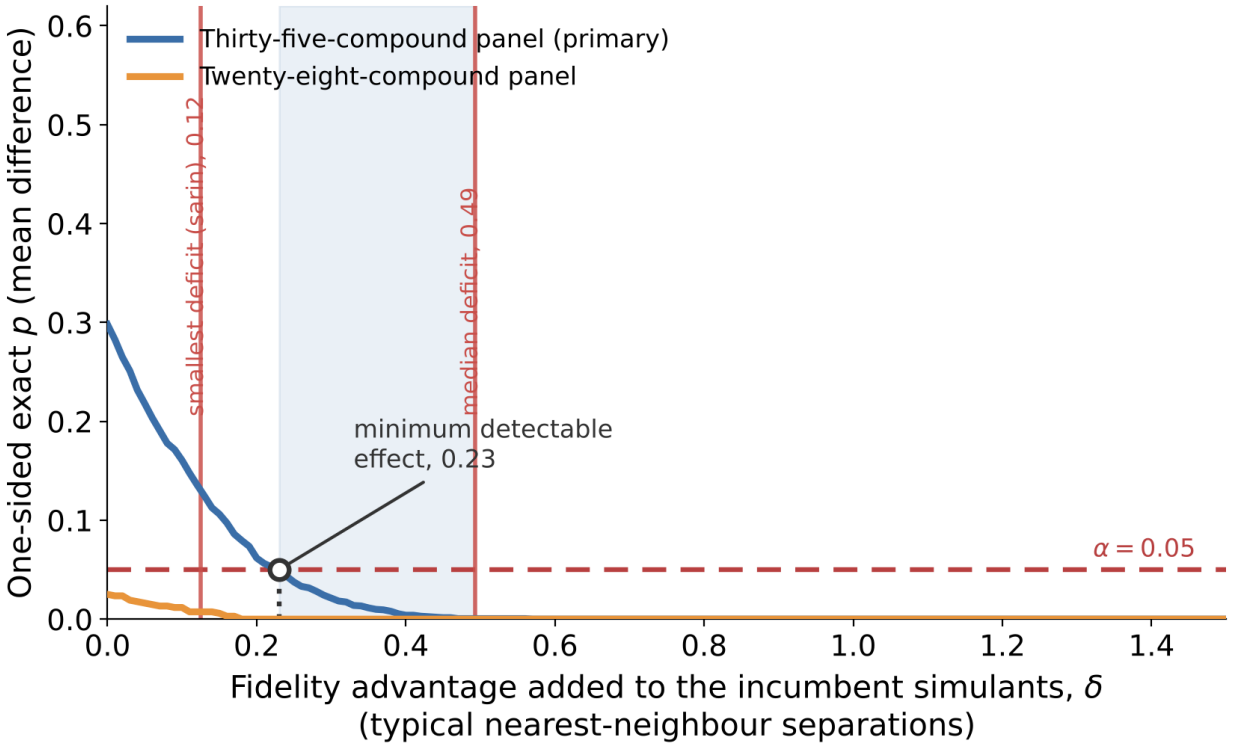



Figure S1: What the permutation test could have detected. One-sided exact $p$ for the mean-difference statistic as the incumbent simulants are moved $\delta$ typical nearest-neighbour separations closer to the eleven agents, with the rest of each panel unchanged and the null taken from the unmodified distances. On the primary panel the test crosses $\alpha = 0.05$ at $\delta = 0.23$. The shaded band spans that value and the median deficit the incumbent compounds actually carry, 0.493; the deficit falls below the detectable value only at sarin.

## S23 Local reactivity at phosphorus

Table S10 gives the phosphorus-centred quantities. The relevant reactivity is the attack of the serine hydroxyl of acetylcholinesterase on phosphorus, so the quantity of interest is the susceptibility of that centre to a nucleophile, which is what $f^{+}(\mathrm{P})$ measures.

$f^{+}(\mathrm{P})$ orders GB > GD > GF > DMMP > GA > VX. DMMP reaches 84% of sarin's value but lies 4.47 standard deviations below the tight GB/GD/GF cluster ($0.6456 \pm 0.0195$), so it is the closest available compound rather than a member of the group. Against VX,

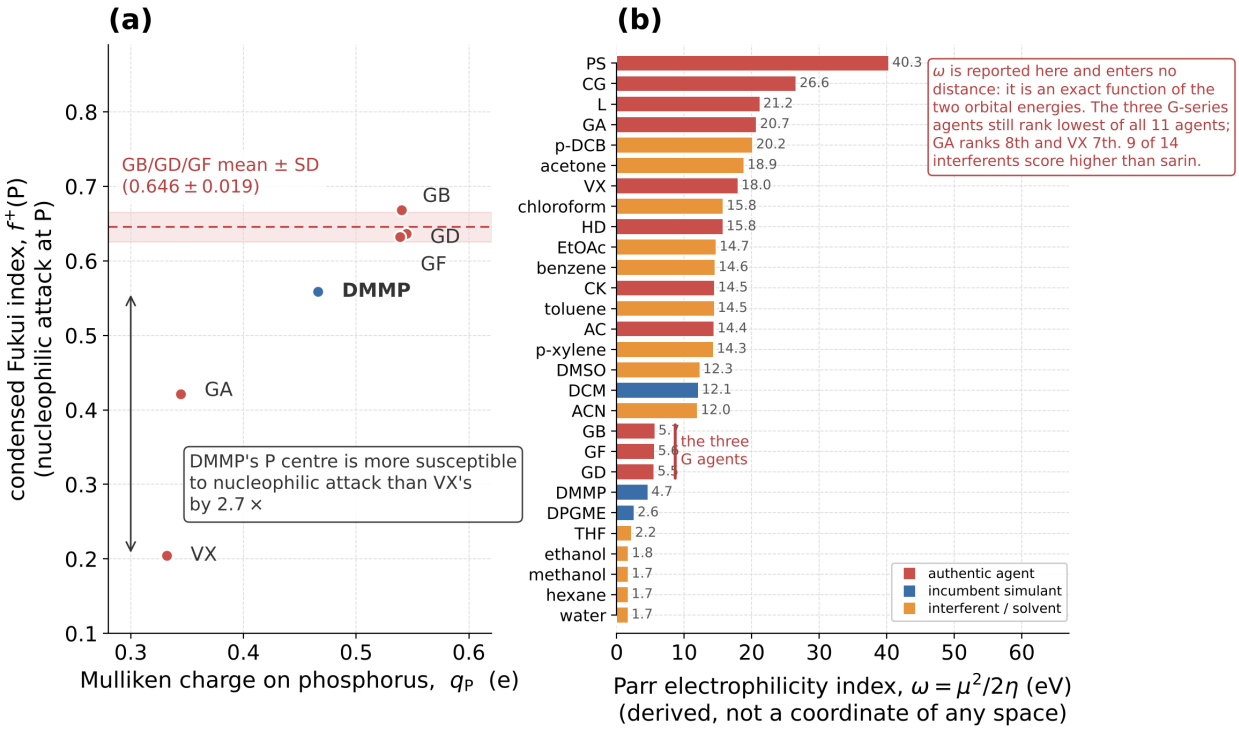



Figure S2: Two quantities that sit outside every distance used in this work. (a) Local electrophilicity against Mulliken charge at phosphorus for the six phosphorus-bearing compounds (Section S23). (b) The Parr electrophilicity index, an exact function of the orbital energies that enter the electronic space, which itself enters no distance anywhere in this work (Section ) and on which the three nerve agents rank lowest of the eleven agents. The principal components of the two spaces are not reproduced here.

DMMP's $f^+$(P) is 2.73 times larger.

That last ratio depends on which quantity is taken to represent local reactivity, and we report both because they disagree. The composite local electrophilicity of Domingo et al.,[S32] $\omega_k = \omega f_k^+$, weights the condensed Fukui index by the global index, and under it DMMP reaches 0.69 of sarin and 0.71 of VX rather than 2.73 times VX: the direction of the DMMP–VX comparison reverses. The reversal is carried entirely by the global factor, which for VX is 17.99 eV because the sulfur lone pair raises the HOMO, and for tabun 20.73 eV because of the nitrile. Those are the same values that the next paragraph shows to be unusable as a reactivity ordering for this compound class, so we do not treat $\omega_k$ as the arbiter; but a reader applying the standard composite to Table S10 would reach the opposite conclusion about VX, and that is worth stating plainly. Against the GB/GD/GF cluster

the two conventions agree and $\omega_k$ is the harsher of the two, placing DMMP 6.87 standard deviations below the cluster against 4.47 for $f^+$(P) alone.

The measured array response bears on this and narrows it. Section S24 compares DMMP's 29-dye response pattern with those of the authentic agents. DMMP is closer to sarin than to VX by 1.35 in cosine in the ultraviolet channel and by 0.84 in reflectance, both intervals excluding zero, so on the VX comparison the measurement sides with $f^+$(P) against $\omega_k$. It does not support the wider reading that DMMP stands in for the G series as a class: GD and GF are both further from DMMP than VX is.

The Wiberg P=O bond order,[S36,S37] which spans 1.9246–2.0146 across all six compounds, does not separate the G series from DMMP; it carries no useful discrimination here. The global electrophilicity index does not order this panel in a way that tracks agent reactivity, and it does not order it reproducibly either. Under GFN2 the three lowest values among the eleven authentic agents are GD (5.54), GF (5.62) and GB (5.69), and nine of the fourteen interferents score higher than sarin. The reason $\omega$ behaves this way is that it responds to the lowest $\pi^*$ orbital wherever it sits, the nitrile in tabun or the sulfur lone pair raising the HOMO in VX, and it has no particular connection to phosphorus.

That much is a statement about $\omega$. Whether it is also a statement about GFN2 is a separate question, and it has to be asked, because the GFN2 values themselves are not credible as electrophilicity indices: they run from 1.71 to 40.34 eV across this panel, where a density-functional calculation on organic molecules of this kind returns single-digit values. We recomputed the whole twenty-eight-compound panel at B3LYP/def2-SVPD (Section S10). The range becomes 1.43 to 7.92 eV, chloropicrin falling from 40.34 to 7.92 and phosgene from 26.60 to 5.07, and the inflation factor is not constant: it runs from 0.7 for water to 11.7 for VX.

Two things follow, and they point in opposite directions for us. The qualitative reading survives. Under the density-functional values the simulant still does not sit above the agents it stands in for, DMMP at 1.90 against sarin at 2.08, and seven of the fourteen interferents

still score above sarin, so $\omega$ still fails to separate this panel in the way a reactivity axis would need to. The specific ordering does not survive. The three lowest agents become VX (1.54), sulfur mustard (1.73) and cyclosarin (1.90) rather than the G series, and two agents now fall below DMMP where under GFN2 all eleven stood above it. The rank correlation between the two sets of $\omega$ values is 0.628, against 0.898 for the underlying LUMO energies, which is what one expects of a quantity that divides by a gap: the construction amplifies the disagreement rather than inheriting it.

We therefore keep the conclusion and withdraw the ordering. Global electrophilicity should not be used as a reactivity axis for this compound class, and the sentence we would have written before this check, naming which three agents sit lowest, would have been a statement about the semi-empirical method rather than about the compounds.

## S24 What the array actually measures

Every argument above rests on descriptor distances, and the premise that one of those spaces corresponds to what a vapour-phase sensor reports has so far been assumed rather than shown. We test it against measured response from a 29-dye colorimetric array, reported for three compound sets at four concentrations in two imaging channels, ultraviolet-excited and reflectance (Section S15). Twelve of the measured compounds are in the panel: DMMP and the five nerve agents in the first set, and six of the remaining agents in the second.

The array discriminates, but only after the common-mode response is removed, and less of that separation is empirical than the raw numbers suggest. Colour difference is a non-negative quantity, so every analyte shares a common "something responded" component and cosine similarity between raw profiles is compressed towards unity. Between-compound similarity of 0.77 to 0.80 is not separable from the within-compound value of 0.79 to 0.83 across concentrations ($p = 0.06$ to 0.94). Subtracting the compound-axis mean at each concentration separates them, to $+0.3451$ to $+0.5416$ within compounds against $-0.1026$ to

$-0.3302$ between them, in all six combinations of compound set and channel.

The sign-flip enumeration we pre-specified for this contrast returns the smallest value each compound count allows in every cell, from 0.0078 to 0.0625, and Section S15 withdraws it. The test reported below permutes compound labels within each concentration. The statistic is unchanged, the mean over compounds of the within-concentration agreement minus the between-compound agreement, and only the null changes.

Table S11 gives the result. All six cells clear $q = 0.05$, the smallest adjusted value being 0.0001, and the null mean is positive in every cell rather than zero. That positive null mean is the quantity the withdrawn floor argument was reaching for: it says how much of the observed difference centring produces on its own, between $+0.1024$ and $+0.3138$ against observed values of $+0.5578$ to $+0.7288$. The reflectance channel of the third run, which the floor comparison discarded for sitting 0.0032 above $-1/3$, has the largest null mean of the six and still clears the threshold.

Two limits stay. The contrast holds per compound on average and not pairwise, and the main text carries that count as its headline. In 13 of the 34 compound-and-cell instances some other compound's consensus is closer than the compound's own agreement across concentrations, and in 28 of the 34 before centring, against a label-permutation expectation of 29.8 and 33.8. Four of the six cells fall below $q = 0.05$ against that null after centring, from under 0.0001 to 0.0120, and three before it; the cells that do not are the two channels of the third run after centring, at 0.139 and 0.151, with the first run's ultraviolet channel at 0.070 joining them before it. The smallest value a cell can attain under this null is 0.0349 or below in every cell and in both representations, so a cell that does not reach 0.05 had the resolution to reach it, and the failure is in the data rather than in the test. Collapsing the duplicate dye channels moves the count after centring to 14 of 34 and leaves the count before it unchanged at 28. The per-cell counts, the nulls with their five seeds, the Benjamini–Hochberg table and the attainable minima are in `outputs/reversal_audit.json`. And the sampled cells give a resolution of $10^{-4}$ in $p$, so the four sampled values are upper bounds on how small those

values are rather than estimates of them. All results below use the centred patterns, with those caveats attached.

The measurement bears on Supplementary Section S23, and it narrows the claim rather than confirming it. What replicates across both channels is specific to sarin. DMMP's pattern is closer to GB than to VX by 1.347 in cosine in the ultraviolet channel (95% CI 1.062 to 1.436, bootstrapped over concentrations) and by 0.841 in reflectance (0.462 to 0.903), and closer to GB than to the mean of GD, GF and VX by 1.284 and 0.984 in the two channels, every interval excluding zero. The class-level statement is weaker. Taking the mean of the three matched G agents against VX gives 0.513 (0.131 to 0.722) in the ultraviolet channel but 0.137 (-0.144 to 0.290) in reflectance, and the least similar agent to DMMP is GF rather than VX in both channels (-0.721 and -0.496 against -0.702 and -0.273 for VX).

The full ranking contains one result that the descriptor analysis does not anticipate, and we give the whole ranking rather than the comparison we set out to make. In the ultraviolet channel the agent nearest to DMMP is not sarin but tabun, at +0.764 against sarin's +0.645, with soman, VX and cyclosarin following at −0.492, −0.702 and −0.721. In reflectance sarin leads at +0.569 and tabun falls to −0.112. Tabun sits on the descriptor criterion of Section rather than inside it: its normalised distance is 2.0038 against a threshold of 2, where sarin sits at 1.187, and Section S6 shows that implementation choices carrying no chemical content move it to either side of that threshold. The descriptor table therefore does not count tabun among the matches, while the measured ultraviolet channel places it first of the five agents. The two do not disagree about a settled count. They disagree about the order, and they disagree most about the one agent the descriptor criterion cannot resolve. The measured channel in which the ordering is clearest puts tabun first, the descriptor table puts it last of the four, and the agreement between the two channels on tabun is the worst of any agent in the set. We report this as a disagreement between the descriptor table and the measurement and not as a correction to either, because two channels on one plate set cannot settle which is right.

The descriptor analysis assigns GB, GD and GF to DMMP and leaves GA on the threshold. The array places sarin first in one channel and tabun first in the other, and does not place GD or GF closer to DMMP than VX in either. A sensor chemistry calibrated on DMMP is therefore being shown something that behaves like one G agent at a time, not like the G series, and the descriptor match for the other two does not survive contact with the measured response.

It does not answer the larger question in either direction. Pattern dissimilarity, taken as $1 - \cos$ and correlated against descriptor distance, gives the same inconsistent picture as the permutation tests of Section . Tested by Mantel permutation over compounds (Section S15), three of the twelve panel-channel-space cells reach a nominal $p < 0.05$, and they do not agree on a space: the merged space in the second compound set's ultraviolet channel ($\rho = +0.500$, Mantel $p = 0.019$), the structural space in that same set and channel ($\rho = +0.511$, Mantel $p = 0.017$), and the electronic space in the same set's reflectance channel ($\rho = +0.571$, Mantel $p = 0.044$). The first compound set gives nothing in any space or channel ($|\rho| \leq 0.225$, Mantel $p \geq 0.22$).

Which set the three nominal cells come from decides what they can mean, and we state it because the composition of the two sets is not symmetric. The first set is DMMP with five G agents and VX, and it is the only one of the two that contains a simulant at all; the second is seven authentic agents and no simulant. A cell in the second set therefore asks whether the pattern distance between two agents tracks their descriptor distance, which is a different question from whether a simulant stands in for an agent. All three nominal cells are in the second set. The six cells that bear directly on the premise this paper rests on are the six in the first set, and all six are null. We have not shown that the descriptor spaces fail to predict simulant-to-agent response; we have shown that the only measurement able to address it does not address it, on six compounds, at a resolution that would have needed a near-deterministic association to register. The twelve cells are one family, and we correct them as we correct the permutation tests of Section : under Benjamini-Hochberg the

smallest adjusted value is 0.117 and no cell reaches $q < 0.05$. We state the nominal values as well because cells pointing at different descriptor spaces is itself the pattern worth seeing, but no cell is a result. An earlier version of this analysis ran the Mantel test in the structural and electronic spaces only and corrected over the resulting eight cells. The merged space is the one the rest of this paper leads with, and the cell it contributes is the strongest of the twelve, so omitting it narrowed the family after the fact and in the direction that weakened the association. Restoring it moves the smallest adjusted value from 0.133 to 0.117.

Not every declared channel is a separate measurement, and in one plate set six of them are not. In the second set's ultraviolet channel, dyes 22 to 27 return values identical to the last decimal place in every row. The images behind them are distinct files, so this is not a naming collision. What the deposited tables establish is the identity, not its cause: the most likely reading is that those regions of interest read plate background rather than a dye response, which would make one background channel enter the unit-normalised pattern six times, but the tables begin at $\Delta E_{00}$ and the source reports no per-channel diagnostic, so we record that as our inference and not as a finding. The consequence for the analysis is the same either way, because six identical columns carry one channel's information whatever produced them. Two further columns in the same channel, and five in the third set's, are identically zero. We collapse columns that are equal in every row to a single occurrence, a rule declared once and applied to all six plate-channel combinations, which leaves 23 effective channels in the second set's ultraviolet data and 25 in the third set's, against 29 elsewhere. The collapse touches the three cells of the second set's ultraviolet channel and weakens the two of them that carry the largest coefficients, the merged space from $\rho = +0.5500$ to $+0.5000$ and the structural space from $+0.5464$ to $+0.5107$, and it moves the smallest adjusted value over the twelve cells from 0.0583 to 0.1167. We report the collapsed values throughout, because a paper arguing that analyst choices inflate results cannot keep a sixfold duplicate that happens to fall in its two strongest cells. An earlier version of this section gave six numbers here that no run reproduces; they are replaced by the measurement, which is in

`outputs/collapse_sensitivity.json` of the deposit together with the columns that were collapsed.

Treating the fifteen pairs as independent observations, which the first version of this analysis did, moves these values by up to 0.038 and in both directions: it would have reported the structural cell at $p = 0.052$ rather than 0.017 and the merged cell at 0.058 rather than 0.019. The error is not large here because the sets are small, but it is the error that would grow with panel size, and in the electronic cell it runs the other way, to 0.026 from 0.044.

Pooling the two compound sets by Fisher transformation, which is legitimate because they are different compounds on different plates, the structural space gives $\rho = +0.306$ (95% CI $-0.450$ to $+0.806$) in the ultraviolet channel and $+0.141$ ($-0.577$ to $+0.736$) in reflectance, while the electronic space gives $-0.107$ ($-0.720$ to $+0.600$) and $+0.327$ ($-0.431$ to $+0.814$). The weight here is $n - 3$ with $n$ the six compounds, not the fifteen pairs they generate; the earlier version of this section used the pairs, which is the error it criticises two paragraphs above and which narrowed every interval by roughly a factor of two. Every interval contains zero. The electronic space is the better predictor in one of the four cells.

Two of the twelve cells are exact duplicates of two others. On the first compound set the correlation filter leaves the merged space identical to the structural space in both channels, so `d1.uv/merged` reproduces `d1.uv/structural` and `d1.normal/merged` reproduces `d1.normal/structural`, to the digit. Counting them separately enlarges the correction family and therefore moves the result towards the null we report, which is the conservative direction. Removing them and correcting over the remaining ten gives a smallest adjusted value of 0.0972 against the 0.1167 we report, and changes no conclusion.

Six compounds per set bound how much that null can carry, and bounding it correctly took us two attempts. A Fisher $z$ calculation on the fifteen pairs returns $|\rho| \geq 0.669$ at 80% power; the same calculation with the six compounds as the unit, which is the unit the Mantel test uses, returns 0.924. Neither is the test we ran. The cell test enumerates all $6! = 720$ relabellings of the compounds, and that permutation null is narrower than one built from

independent pairs, because it preserves the dependence among the fifteen pairs rather than ignoring it. A narrow null compensates for a small sample, so the approximation at the correct unit understates the power of the procedure. We therefore simulated the procedure itself (Section S15): at $\alpha = 0.05$, 80% power is reached at $|\rho| = 0.67$, with the type I rate of the simulated test at 0.05 against a nominal 0.05, measured on two independent configurations over 20,000 null replicates with a standard error of 0.0015. That figure is for a single test at the nominal level, and it is not the threshold the conclusions use. The twelve cells are corrected by Benjamini–Hochberg, so the smallest-ranked cell has to clear $0.05/12 = 0.00417$, and at that threshold the same simulation puts 80% power at $|\rho| = 0.85$. We quote both, and read the null against 0.85, for the same reason the designation test is read against its corrected displacement in Section S22 rather than its nominal one. The smallest attainable $p$ is $1/720$, so the resolution is coarse but not as coarse as the compound-unit approximation suggests. These data exclude a strong association between either descriptor space and the measured response and nothing weaker. They do not show that the electronic space governs sensor response, do not place it above the structural space, and do not show it to be irrelevant. The claims in the rest of this paper are written to that limit.

## S25 Two readings that did not survive their own controls

Two further readings suggested themselves and did not survive their own controls. We report both, because the alternative is to present artefacts as results. The first was a ranking. DMMP is the nearest of the five candidates to tabun in the ultraviolet channel and to sarin in the reflectance channel and to nothing else, which reads as evidence that the optical channel decides which agent the standard simulant stands in for. It is not. With five candidates the expected number of first places over ten agent-channel cells is 2.0, the observed number is two, and the one-sided binomial $p$ is 0.62. The ten cells share DMMP's two response vectors, so they are not ten independent trials and the binomial is an approximation; the observed

count sits on the expectation, where no correction could move the reading.

The second was a channel decomposition. The cosine is a sum over dye channels, and the DMMP-to-tabun value of +0.764 turns out to rest on two of the 29 channels, with the largest single channel carrying 44% of the positive mass. Read alone, that makes the resemblance look fragile. Two controls show it is not a property of DMMP. Agent-to-agent pairs in the same run concentrate identically, with median two channels in the ultraviolet and one in the reflectance channel, and a null built by permuting the dye labels of the DMMP vector gives a mean of 2.00 and 2.26. Concentration of this kind is what the product of two centred unit vectors in 29 dimensions does, and it says nothing about this compound.

Neither reading enters any conclusion. They are set out here so that the record of what was looked at is complete.

## S26 What the dose direction does and does not carry

Three limits sit on that direction. The concentrations are of analyte in a dye solution, not of vapour at an inlet, so we do not translate the range into the operating band of a point detector. The mean response magnitude rises with concentration at a Spearman correlation of exactly +1.000 in every one of the six run-channel cells, so a similarity that grows with concentration cannot be separated here from one that grows with signal-to-noise ratio; distinguishing them needs a repeat-level noise estimate the deposited tables do not carry.

### Two readings of the simulant cell

A weaker claim survives in one channel. If DMMP is G-like at all it should sit closer to the G series than to VX, and that contrast, against the mean over the three matched G agents and excluding tabun, is +0.513 in the ultraviolet channel and +0.137 in reflectance. The ranges quoted with them, +0.131 to +0.722 and −0.144 to +0.290, come from resampling the four

design concentrations with replacement; four fixed design levels are not an exchangeable sample, so these are descriptive ranges and not tests, and the ultraviolet range lies above zero while the reflectance range spans it.

Concentration changes the picture (Figure 2c of the main article). The DMMP-to-sarin cosine rises monotonically from +0.363 at 10 $\mu$M to +0.771 at 500 $\mu$M, and the DMMP-to-cyclosarin cosine falls monotonically from $-0.462$ to $-0.700$ over the same range, with monotonicity perfect in three of the ten agent-channel series. That is also the strongest result the design can give: with four concentrations the two-sided exact permutation has a floor of $p = 0.083$, three series reach exactly that floor, and Benjamini–Hochberg correction over the ten leaves a smallest adjusted value of 0.278. We report the direction and not a significance claim.

What the direction supports is that the resemblance between this simulant and this agent is not constant over the range at which the array was characterised. Two limits sit on it, the solution-phase concentration axis and a signal-to-noise confound that these tables cannot separate from it, and both are stated above.

And the agent stock of the first run was 1 M in tetrahydrofuran, diluted with acetonitrile, so the residual tetrahydrofuran in the vial is proportional to the final analyte concentration, rising from roughly 0.001 to 0.05 per cent by volume across the four levels. A solvent gradient of that shape is a third candidate for anything that varies monotonically with dose, and these tables cannot separate it from the other two either. What the direction supports is that the resemblance between this simulant and this agent is not constant over the range at which the array was characterised.

These tables hold one value per compound, concentration, dye and channel, so the within-compound term is agreement of a compound with itself at a different dose and not between repeated measurements. Repeats were reported in the source work for selected dye-analyte combinations in the second run[S22] and for all experiments in the third,[S24] but the repeat-level values are not in these tables, so no repeatability estimate enters any analysis below.

# S27 The measured safety axis, and the holes in it

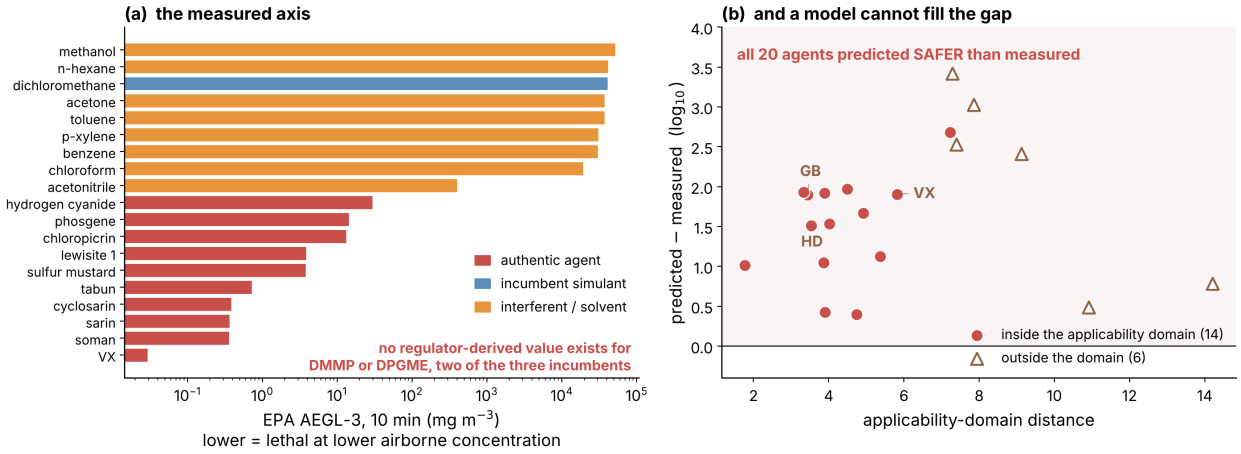



Figure S3: The toxicity axis. (a) Measured EPA AEGL-3 10-minute values for the panel compounds that carry one. (b) Prediction error of a random forest trained on 230 AEGL chemicals, for twenty military agents held out of training, against distance to the training set.

Nineteen of the twenty-eight compounds have an AEGL-3 at 10 min (Fig. S3a). The ordering runs VX (0.0295 mg m$^{-3}$), soman (0.365), sarin (0.367), cyclosarin (0.391), tabun (0.729), sulfur mustard (3.84), lewisite 1 (3.9), chloropicrin (13.4), phosgene (14.6), hydrogen cyanide (29.8), acetonitrile (403), chloroform (19 530), benzene (30 990), p-xylene (31 264), toluene (37 685), acetone (38 007), dichloromethane (41 685), n-hexane (42 296) and methanol (52 420). The axis spans six orders of magnitude, and the incumbent simulant dichloromethane sits at the safe end of it, $1.4 \times 10^6$ times less acutely toxic than VX and 1400 times less toxic than hydrogen cyanide. On this axis the incumbent simulant is not the weak link.

It is, however, not the safest thing on the panel. Methanol and n-hexane both carry higher AEGL-3 values than dichloromethane, and neither is used as a simulant for anything. The point is not that either should be: it is that the safety axis cannot be what selected the incumbent, because two ordinary laboratory solvents on the same panel score better on it. Against the 207 AEGL compounds that carry both axes (Section S30) dichloromethane sits at the 96th percentile of measured acute inhalation safety, with seven compounds above it.

Taking the incumbent as the bar a replacement must meet therefore sets a demanding bar, and the consequences are worked through in Section S30.

Nine compounds have no value. Eight are absent from the AEGL list altogether: DMMP, tetrahydrofuran, ethyl acetate, water, DPGME, ethanol, DMSO and p-dichlorobenzene. The ninth, cyanogen chloride, is the one military agent among them, and its absence is a recorded decision rather than an oversight: it appears in the compiled table's holding list, where the agency states that values are not being developed. Checked by CAS against the source document, it is the only one of the nine on that list. DMMP's position is worth stating precisely, because it is the compound that carries all three valid matches found above. EPA's provisional peer-reviewed assessment declined to derive an inhalation reference concentration, recording that the data are insufficient, and derived no inhalation unit risk; no acute inhalation $LC_{50}$ is cited; IRIS and HEAST contain no RfD, RfC or cancer value. What exists is an oral provisional RfD of 0.06 mg $kg^{-1}$ $day^{-1}$ carrying an uncertainty factor of 3000, and an oral cancer slope factor of $1.7 \times 10^{-3}$ (mg $kg^{-1}$ $day^{-1})^{-1}$ based on mononuclear cell leukaemia in male rats, with the evidence characterised as suggestive.[S38] The inhalation safety of the field's principal nerve-agent simulant has not been established; it has not been assessed. The two incumbent simulants therefore sit at opposite ends of the evidence, not of the hazard: for dichloromethane there is a regulator-derived value and it is favourable, and for DMMP there is a regulator's recorded decision not to derive one.

Cyclosarin carries its own entry in the compiled table under CAS 329-99-7, with a ten-minute AEGL-3 of 0.053 ppm bracketed by the agency as 0.38 mg $m^{-3}$; our conversion from the published parts-per-million value gives 0.39 mg $m^{-3}$, placing it between sarin and tabun. The entry is titled *GF Agent* rather than *Agent GF*, which is why matching on compound names rather than on registry numbers can miss it; Section S1 describes the CAS-keyed procedure used here.

## S28 What each register holds for the three incumbents

The toxicity axis of the main text is the EPA acute exposure guideline level, because that is the register on which the eleven agents are placed. Two of the three incumbents are absent from it. They are not absent from every register, and an earlier version of this work said that they were; the correction is recorded here with its sources.

Table S12 lists what we found. The US Department of Energy protective action criteria hold values for DMMP, derived as temporary emergency exposure limits from a rat one-hour $LC_{50}$ rather than from a human-health assessment, and NIOSH holds an immediately dangerous to life or health value and a recommended exposure limit for DPGME. These are real regulator-derived numbers and we do not discount them. They are also not the quantity the agents carry. An AEGL-3 is a ten-minute, three-tier, lethality threshold set by a federal advisory committee from a documented weight of evidence; a PAC-3 is a sixty-minute value that falls back to a formula on an animal $LC_{50}$ when no AEGL or ERPG exists; an IDLH is a thirty-minute escape criterion. Placing a PAC-3 for the simulant beside an AEGL-3 for the agent would put two different quantities on one axis and report the difference between them as a difference between compounds.

A single-register axis that reaches more of the panel would be the better instrument, and the protective action criteria are the obvious candidate: they cover compounds the AEGL list does not, including DMMP and cyanogen chloride. We did not build it. The values have to come from the register's own tables and we were not able to obtain them for every panel compound, so a PAC axis here would have had worse coverage than the AEGL axis it replaced. We record it as the measurement this comparison still needs.

## S29 A QSAR cannot fill the gaps

On the training distribution the model works. The random forest (Fig. S3b) reached $R^2 = 0.458 \pm 0.014$ and RMSE 0.935 in $\log_{10}$ units (95% CI 0.830–1.020) against a training-mean

baseline of 1.271 (1.175–1.365) and a logP-plus-molecular-weight linear baseline of 1.225. The intervals for the model and the mean baseline do not overlap, and the out-of-fold error over the whole training set is unbiased, averaging +0.001 log units. Nothing in the cross-validated statistics would warn a user off.

On the twenty agents held out of training it failed, in one direction and without exception. The mean error over the twenty is +1.68 log units, with Cohen's $d_z = +1.96$ over the held-out set. Every one was predicted safer than measured, by between 0.40 and 3.42 log units, a factor of 2.5 to 2600. Sarin was placed at 29 mg m$^{-3}$ against a measured 0.367 and VX at 2.4 against 0.0295, errors of 79-fold and 80-fold. Fourteen of the twenty lie inside the applicability domain, so for those the domain check raises no objection, and the worst of those fourteen is still wrong by 2.68 log units, a factor of 482. DMMP's domain distance, 3.28, sits among the agents' own, which is to say inside the region where the model is wrong by one to two orders of magnitude, so its prediction is not used.

The explanation is not that the mechanism is missing from the training set. We checked, because that was our first hypothesis and it is wrong. Thirteen of the 230 training compounds carry an acetylcholinesterase-inhibiting pharmacophore, identified by substructure rather than by name: ten organophosphorus esters (phorate, phosphamidon, monocrotophos, dicrotophos, fenamiphos, parathion, methyl parathion, methamidophos, dichlorvos, malathion) and three carbamate esters (aldicarb, oxamyl, methomyl). The mechanism by which the nerve agents kill, phosphylation of the serine hydroxyl of acetylcholinesterase and the cholinergic crisis that follows,[S39–S41] is therefore represented, and one of the thirteen, phorate at 0.22 mg m$^{-3}$, is more acutely toxic than sarin. The model sees the mechanism and still cannot place the agents.

What it cannot do is reach either extreme. Over the whole training set the out-of-fold error is centred on zero, but in the most toxic decile (the 23 compounds at or below 3.21 mg m$^{-3}$) it averages +1.41 log units and is positive for every single compound, while in the least toxic decile (the 23 at or above 8094 mg m$^{-3}$) it averages −1.13 log units and

is negative for all but one. The same symmetry appears inside the anticholinesterase subset taken on its own: phorate, the most toxic of the thirteen, is predicted 177 times too safe, while malathion, the least toxic, is predicted 269 times too toxic. This is regression toward the mean at both tails, which is what a variance-minimising ensemble does with sparsely populated extremes, and the agents are all in one of them.

That diagnosis matters for the applicability domain rather than for the model. A descriptor-distance domain asks whether a compound is unusual in the space of inputs. The failure here is defined in the space of the response, and the two do not coincide: fourteen of twenty agents are ordinary-looking inputs with extreme outputs. An input-space criterion cannot detect an output-space bias, so it certifies exactly the predictions that are worst. This is the same failure, in the safety axis, that Section  documents in the fidelity axis. A distance in one space is being used to vouch for a property that lives in another.

## S30 Fidelity and safety are not monotonically associated

Across the 207 candidates carrying both measured axes (Fig. S4a), fidelity and safety are uncorrelated: Spearman $r = +0.062$ (95% CI $-0.077$ to $+0.199$), Pearson $r = +0.072$ (95% CI $-0.092$ to $+0.225$, $p = 0.30$), with fidelity accounting for 0.5% of the variance in toxicity (Table S13). The Spearman interval rules out any monotone association stronger than $|r| = 0.199$ and the Pearson interval any linear association stronger than $|r| = 0.225$, so this is a bounded negative result rather than an underpowered one. Resembling an agent electronically does not require being as dangerous as one.

Setting the safety bar at the incumbent simulant's own measured value empties the faithful-and-safe quadrant outright: of the 207, not one compound combines top-quartile fidelity with an AEGL-3 at or above dichloromethane's 41 685 mg m$^{-3}$, and dichloromethane does not either, its own fidelity of 0.78 falling outside the 0.71 quartile boundary. That is a statement about how safe dichloromethane is, not about how scarce candidates are. Eight

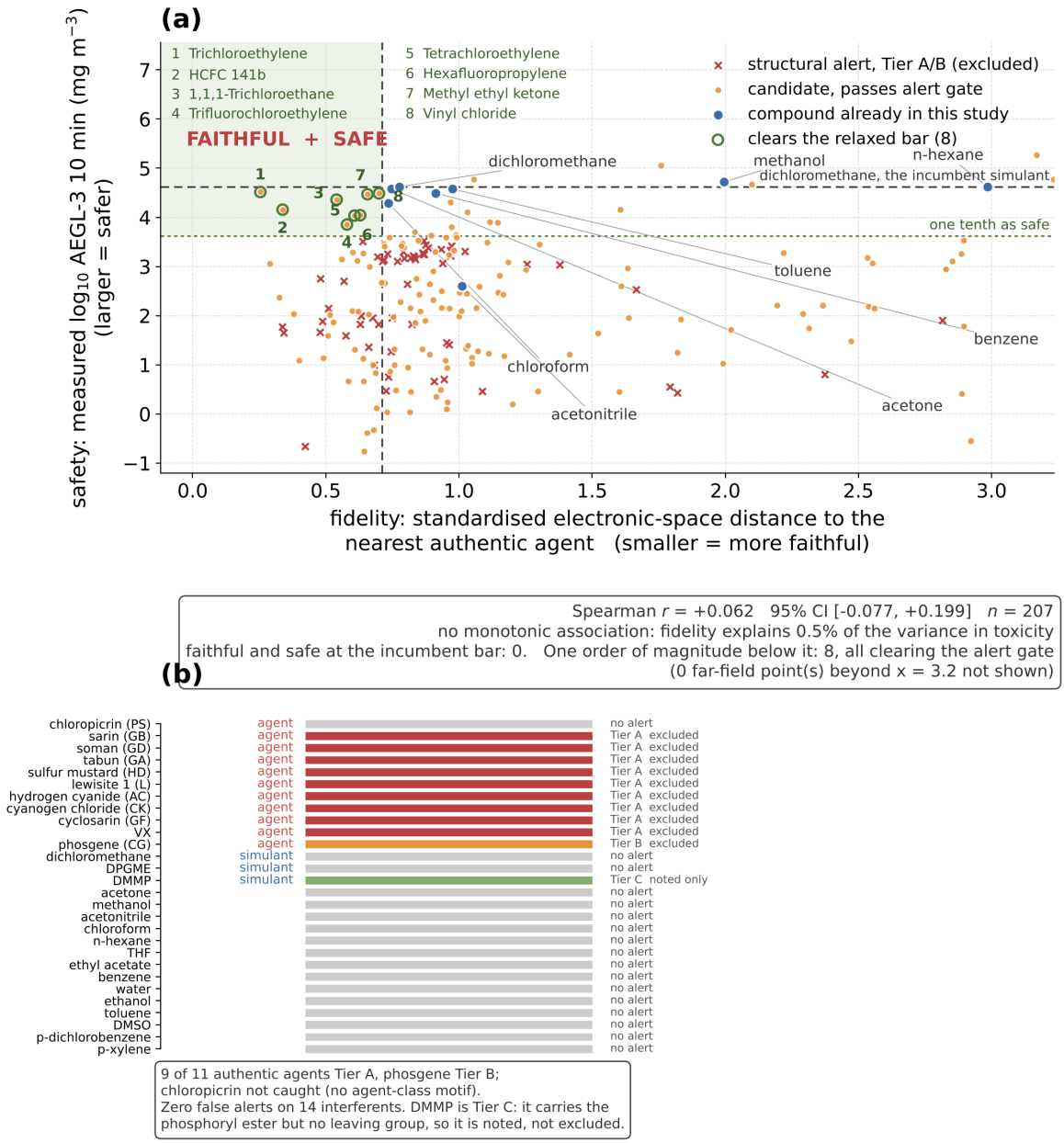



Figure S4: Fidelity and safety are not monotonically associated. (a) Electronic-space distance to the nearest authentic agent against measured AEGL-3 for the 207 candidates carrying both axes; crosses are compounds excluded by the structural filter. The upper dashed line is the measured AEGL-3 of dichloromethane, the incumbent simulant; the dotted line is one order of magnitude below it, and the eight circled compounds are those that clear the lower bar while also falling in the most faithful quartile, numbered by fidelity and named at the top left. The shaded quadrant, faithful and safe at the incumbent's own bar, is empty. (b) The filter applied to the twenty-eight-compound set: nine of eleven authentic agents are Tier A and phosgene Tier B, there are no false alerts among the interferents, and DMMP is Tier C.

compounds in the whole pool reach or exceed it on the safety axis, the incumbent among them, and not one of the eight is faithful.

Lowering the bar by one order of magnitude, to 4168 mg $m^{-3}$, changes the answer. Eight compounds then combine top-quartile fidelity with that level of measured acute inhalation safety, every one of them clears the structural-alert gate, and every one of them is more faithful than dichloromethane itself. In order of fidelity they are trichloroethylene (0.25), HCFC 141b (0.34), 1,1,1-trichloroethane (0.54), trifluorochloroethylene (0.58), tetrachloroethylene (0.61), hexafluoropropylene (0.63), methyl ethyl ketone (0.66) and vinyl chloride (0.70), every one of them ahead of dichloromethane's own 0.78. The best of them, trichloroethylene, is 3.1 times closer to the nearest authentic agent in the electronic space while being only 1.3 times more acutely toxic by inhalation. At two orders of magnitude below the incumbent the count rises to fifteen, all of which clear the gate.

These are candidates on the axes measured here, and that qualification is not a formality. Trichloroethylene and tetrachloroethylene carry serious chronic and carcinogenic hazard that an acute inhalation criterion does not express, and the chlorofluorocarbons among them raise questions of atmospheric persistence rather than of laboratory safety. What the screen establishes is narrower and still useful: in the electronic space, compounds more faithful than the incumbent exist at comparable acute inhalation safety, and they were not found by relaxing the chemistry but by looking in the right descriptor space.

One limitation bounds this part of the result. The candidate pool is the set of chemicals for which AEGL values exist, and chemicals receive AEGLs because they are hazardous. Ordinary safe solvents are largely absent from the pool by construction, which depresses the count at any bar. The seven compounds that exceed dichloromethane on the safety axis illustrate the point from the other side: two are hydrocarbon gases, one a refrigerant, one a monomer, and the three a laboratory might actually reach for sit far out on the unfaithful side of the electronic axis, methanol at 2.00 against the 0.71 boundary, well outside the top quartile that defines a faithful candidate.

## S31 The structural filter

Applied to the 28-compound set (Fig. S4b), the filter assigns Tier A to nine of the eleven authentic agents and Tier B to phosgene, catching ten of eleven; chloropicrin carries no agent-class motif and is not caught. There are no false alerts among the fourteen interferents. DMMP is Tier C: it carries the phosphoryl ester but not a leaving group, which is precisely why it is used as a simulant and why it is noted rather than excluded. Dichloromethane and DPGME raise no alert.

The five compounds the primary panel adds have not been put through the filter anywhere else in this work, and two of them fail it. Diisopropyl fluorophosphate and diethyl cyanophosphonate carry `A_OP_leaving_F_CN`, the same Tier A alert that catches sarin, soman and cyclosarin; diethyl chlorophosphate is Tier C with the phosphoryl ester alone, and methyl salicylate raises nothing. These are the compounds that displace the incumbents on the fidelity axis, so the two axes of the criterion point in opposite directions for them, and nothing here should be read as proposing either as a replacement. 2-Chloroethyl ethyl sulfide raises no alert at all, which is a false negative rather than a finding: `A_sulfur_mustard` requires a 2-haloethyl group on both arms of the sulfide, so half-mustards pass a filter written for the agent. We record it beside chloropicrin as a second known gap in the filter's coverage. None of the five carries a measured acute inhalation value in the panel assembled here, so the safety axis is unevaluated for all of them.

## S32 Further discussion of the toxicity axis

The decoupling result is what makes the rest actionable. Because fidelity and acute toxicity show no monotonic association across the compounds where both are measured, the pursuit of a simulant that is both faithful and safe is not a pursuit along a trade-off curve, and the screen bears that out. Against the incumbent's own safety value the quadrant is empty, which says more about dichloromethane than about the chemistry: only seven of the 207

candidate compounds are safer than it, and it is not itself faithful enough to fill the quadrant it empties. One order of magnitude below that bar a different count applies: there are eight compounds, all of them clearing the structural-alert gate and all of them more faithful than the incumbent, with trichloroethylene 3.1 times closer to the nearest agent in the electronic space at 1.3 times the acute inhalation toxicity.

We are not proposing trichloroethylene as a simulant. It carries chronic and carcinogenic hazard that the axis used here does not express, and a replacement chosen on two measured axes still has to survive the others. The claim is narrower and, we think, more useful: the field's incumbent choice is not on the efficient frontier of the two axes it is usually defended on, and better points on those axes are already present among compounds with measured toxicity. That the pool consists of chemicals assessed because they are hazardous depresses every count reported here, so both counts are lower bounds rather than inventories. The constructive reading is that designing candidates and then measuring them remains worthwhile, and that the filter excluding agent-class motifs by construction can be applied from the start.

**Limitations specific to this material.** The safety axis is acute inhalation lethality at ten minutes and does not represent chronic, dermal or ocular hazard, nor flammability, reactivity or environmental persistence, which is the main reason the eight candidates of Section S30 are presented as positions on two axes rather than as recommendations. Four phosphorus-local descriptors exist for six compounds only and are reported without being used in any 28-compound analysis. Two features of the regulatory source are carried into our numbers and should be read with them. Five of the nineteen panel values, and nine of the 259 in the wider set, are the agency's footnote values for exposures above half the lower explosive limit; they are the agency's ten-minute AEGL-3 for those compounds, but at concentrations where explosion rather than toxicity is the governing hazard, which is a constraint on using them as working exposure limits even though it does not affect their use

here as a relative toxicity scale. And the panel's p-xylene is matched to the compiled table's *Xylenes* entry (CAS 1330-20-7), which covers the mixed isomers rather than the para isomer specifically; p-xylene's descriptors are its own, but its safety-axis value is a class value.

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Table S2: Every simulant designation we located, with the agent it names, the measurement it was made against, and the basis the designating document itself states. Source URLs and the search date are in `sources/designations_20261106.csv` of the deposit.

| Compound | Document | Agent | Measurement | Basis the document states |
|---|---|---|---|---|
| DMMP | NIOSH CBRN APR simulant programme (31 July 2002, draft for discussion) | GB and HD | filtration | relatively non-toxic; easily detectable; volatility intermediate between GB and HD |
| carbon tetrachloride | NIOSH CBRN APR simulant programme (31 July 2002, draft for discussion) | GB and HD | filtration | named with cyclohexane as a GB and HD filtration simulant; no separate basis stated |
| cyclohexane | NIOSH CBRN APR simulant programme (31 July 2002, draft for discussion) | GB and HD | filtration | named with carbon tetrachloride as a GB and HD filtration simulant; no separate basis stated |
| DMMP | NRC, Technical Assessment of the Man-in-Simulant Test Program (1997), ch. 3 | VX | fabric penetration | low toxicity and similarity of certain physical properties; chemical structure may be quite different |
| DIMP | NRC, Technical Assessment of the Man-in-Simulant Test Program (1997), ch. 3 | GB | fabric penetration | low toxicity and similarity of certain physical properties |
| methyl salicylate | NRC, Technical Assessment of the Man-in-Simulant Test Program (1997), ch. 3 | HD | fabric penetration | low toxicity and similarity of certain physical properties |
| DPGME | US Patent 7129094 B1, chemical agent simulant training composition (2006) | [not obtained: patent text not retrieved] | [not obtained: patent text not retrieved] | [not obtained: patent text not retrieved; Google Patents, Justia and the USPTO full-text service all refused; bibliographic record only] |
| dichloromethane | [not obtained: no designating document found in the public record] | none found | none found | none found |

Table S3: The 29-component array, in the order the three source studies number it. Five entries marked [a] are not chromogenic dyes; [b] marks a component whose stock solvent differs between the source studies.

| # | Compound | # | Compound |
|---|---|---|---|
| 1 | Anthracene[a] | 16 | Bromophenol Blue |
| 2 | Pyrene[a] | 17 | Neutral Red |
| 3 | Allura Red AC | 18 | Eosin Y |
| 4 | Quinine | 19 | Indigo Carmine |
| 5 | Rhodamine B | 20 | Phenol Red |
| 6 | Methyl Blue | 21 | Crystal Violet |
| 7 | Fluorescein | 22 | L-Glutathione (reduced)[a] |
| 8 | Methyl Orange | 23 | 2,7-Dibromofluorene |
| 9 | Nile Red | 24 | Ethyl viologen dibromide |
| 10 | Safranin O | 25 | 4,4'-Dihydroxybenzophenone |
| 11 | Toluidine Blue O | 26 | 2,4-Dinitrophenylhydrazine[a] |
| 12 | Nile Blue A | 27 | 2,5-Diaminobenzene-1,4-dithiol dihydrochloride |
| 13 | Cresol Red | 28 | 2-Hydrazinobenzothiazole[a] |
| 14 | Eriochrome Black T | 29 | Rhodamine 6G |
| 15 | Quinoline Yellow[b] | | |

[a] 1 Anthracene: fluorophore rather than a chromogenic dye

[a] 2 Pyrene: fluorophore rather than a chromogenic dye

[b] 15 Quinoline Yellow: stock solvent differs between sources: DMSO in the first and second runs, water in the third. Reported as published; not resolved here.

[a] 22 L-Glutathione (reduced): thiol reducing agent rather than a dye

[a] 26 2,4-Dinitrophenylhydrazine: carbonyl derivatising reagent rather than a dye

[a] 28 2-Hydrazinobenzothiazole: carbonyl derivatising reagent rather than a dye

Table S4: Every exact permutation test across panel constructions, descriptor spaces and statistics. Values are one-sided $p$. Bold marks $p < 0.05$ before correction. The last column is the smallest Benjamini–Hochberg value over the eighteen tests of that panel construction. Dimensions are the retained coordinate counts after the correlation filter of Section S6.

| Panel / simulant set | Space | Dim. | $\Delta\bar{d}$ | $\Delta\tilde{d}$ | $\overline{\ln r}$ | $\bar{d}_{\text{sim}}$ | $n_{\text{closer}}$ | $n_{\text{valid}}$ | |
|---|---|---|---|---|---|---|---|---|---|
| **35 cpd., incumbent 3** | structural | 38 | 0.3508 | 0.3261 | 0.42 | 0.2569 | 1 | 1 | |
| **(primary)** | electronic | 3 | 0.3859 | 0.3735 | 0.4116 | 0.3439 | 0.7233 | 0.664 | |
| | merged | 41 | 0.2979 | 0.2465 | 0.3735 | 0.1917 | 1 | 1 | 0 |
| 35 cpd., eight designated | structural | 38 | 0.06869 | 0.2986 | **0.04966** | 0.1578 | 0.3191 | 0.1594 | |
| (3 incumbent + 5 literature) | electronic | 3 | 0.1215 | 0.08624 | 0.1232 | 0.1271 | 0.1651 | 0.07991 | |
| | merged | 41 | 0.05826 | 0.3064 | **0.03864** | 0.1276 | 0.3191 | 0.1594 | 0 |
| 28 cpd., incumbent 3 | structural | 35 | **0.02794** | 0.1103 | 0.1132 | 0.05441 | 0.1103 | 0.2162 | |
| (omits the seven) | electronic | 3 | 0.275 | 0.175 | 0.2926 | 0.2353 | 0.7574 | 0.7574 | |
| | merged | 38 | **0.025** | **0.04265** | 0.05147 | **0.03382** | 0.1103 | 0.2353 | 0 |
| 35 cpd., ten designated | structural | 38 | **0.01149** | **<0.001** | 0.05495 | 0.2144 | 0.3176 | 0.2678 | |
| (3 incumbent + 5 literature | electronic | 3 | 0.08917 | 0.05262 | 0.05889 | 0.08556 | 0.06638 | 0.05495 | |
| + 2 pesticides) | merged | 41 | **0.004756** | **<0.001** | **0.03517** | 0.1435 | 0.3176 | 0.2678 | < |

Statistics, in order: mean and median of $d_{\text{sim}} - d_{\text{intf}}$; mean $\ln(d_{\text{sim}}/d_{\text{intf}})$; mean $d_{\text{sim}}$; count of agents with a simulant closer than every interferent; count of valid matches.

Table S5: Six ways of accounting for the search, applied to the eighteen tests of each panel configuration. The Westfall-Young min-$P$ value[S26] is exact for the dependence among the tests, since all eighteen share one enumeration of designations; $m_{\text{eff}}$ approximates that dependence from the eigenvalues of the correlation matrix of the eighteen nulls,[S27] used as a Bonferroni divisor. On the primary panel no scheme returns anything below 0.05 and on the smaller panel only the pre-specified-statistic scheme does, while the fourth configuration, which differs from the second by two class labels, is below 0.05 under every scheme.

| Accounting scheme | 35 cpd., incumbent 3 (primary) | 35 cpd., eight designated | 28 cpd., incumbent 3 (omits the seven) | 35 cpd., ten designated (pesticides in) |
|---|---|---|---|---|
| Smallest nominal $p$ (18 tests) | 0.192 | **0.039** | **0.025** | **<0.001** |
| Benjamini-Hochberg over 18 | 0.630 | 0.212 | 0.163 | **<0.001** |
| Benjamini-Hochberg over 6, per space | 0.560 | 0.153 | 0.077 | **<0.001** |
| Westfall-Young min-$P$ over 18 | 0.457 | 0.112 | 0.079 | **<0.001** |
| $m_{\text{eff}}$ of 18 (Li-Ji) | 6.00 | 5.00 | 5.00 | 5.00 |
| Bonferroni at $m_{\text{eff}}$ | 1.000 | 0.193 | 0.125 | **<0.001** |
| Statistic specified in advance (3 tests) | 0.386 | 0.103 | **0.042** | **0.014** |
| Benjamini-Hochberg over all 72 | 0.354 | 0.236 | 0.236 | **0.003** |

Table S6: Adopted vapour pressures for the twenty-eight-compound panel, in descending order. Values are measurements retrieved from PubChem, which carries them from the Hazardous Substances Data Bank and from the government compendia named in the source column; the adoption rule and the tier definitions are in the text. No value was converted between temperatures.

| Compound | $P$ (mmHg) | $T$ (°C) | Tier | Source |
|---|---|---|---|---|
| phosgene (CG) | 1420 | | T1 | Daubert, T.E., R.P. Danner |
| cyanogen chloride (CK) | 1230 | | T1 | Daubert, T.E., R.P. Danner |
| hydrogen cyanide (AC) | 742 | | T1 | Daubert, T.E., R.P. Danner |
| dichloromethane | 435 | | T1 | Boublik/Fried/Hala |
| acetone | 231 | | T1 | PMID:7676461 |
| chloroform | 197 | | T1 | Boublik/Fried/Hala |
| THF | 162 | | T1 | Daubert, T.E., R.P. Danner |
| n-hexane | 153 | | T1 | Daubert, T.E., R.P. Danner |
| methanol | 127 | | T1 | Boublik/Fried/Hala |
| benzene | 94.8 | | T1 | Daubert, T.E., R.P. Danner |
| ethyl acetate | 93.2 | | T1 | Daubert TE, Danner RP |
| acetonitrile | 88.8 | | T1 | Boublik/Fried/Hala |
| ethanol | 59.3 | | TX | Daubert, T.E., R.P. Danner |
| toluene | 28.4 | | T1 | Daubert, T.E., R.P. Danner |
| chloropicrin (PS) | 24 | | T1 | Tomlin CDS, ed. Chloropicrin (76-06-2) |
| water | 23.75 | | T2 | Haz-Map, Information on Hazardous Chemicals |
| p-xylene | 8.84 | | T1 | Chao J et al; J Phys Chem Ref Data 12: 1033 |
| sarin (GB) | 2.86 | | T1 | Epstein J; J Amer Water Works Assoc Jan: 31-. |
| p-dichlorobenzene | 1.74 | | T1 | Daubert, T.E., R.P. Danner |
| DMMP | 0.962 | | T1 | Boublik, T., Fried, V., and Hala, E., The Va. |
| DMSO | 0.6 | | T1 | Daubert TE, Danner RP |
| lewisite 1 (L) | 0.58 | | T1 | Munro NB et al; Environ Health Perspectives |
| soman (GD) | 0.41 | | T1 | Balboa A et al; Vapor Pressure of GD, ECBC-TR |
| DPGME | 0.4 | 26 | T3 | Browning, E. Toxicity and Metabolism of Indu. |
| sulfur mustard (HD) | 0.11 | | T1 | PMID:10585900 |
| tabun (GA) | 0.07 | | T1 | Munro NB et al; Environ Health Perspectives |
| cyclosarin (GF) | 0.044 | 20 | T3 | Augerson W; A Review of the Scientific Liter. |
| VX | 0.000878 | | T1 | Buchanan JH et al; Vapor Pressure of VX, ECB. |

Blank temperature means 25 °C. Tier 1 is a vapour-pressure measurement or compilation at 25 °C, tier 2 a government compendium at 25 °C, tier 3 the nearest available temperature. Source strings are as returned by PubChem; the full strings and every candidate value are in the deposited JSON. Tier X is a compilation value without a stated measurement temperature, used only for solvents. Over this compound class the Clausius–Clapeyron slope is of order 0.03 to 0.05 decades per °C, so a 5 °C mismatch moves a value by roughly 0.15 to 0.25 decades. No value was converted between temperatures.

Table S7: What each agent is nearest to on the volatility axis. The series column is the agent's own class. For three of the eleven agents the nearest compound in the pool is an interferent rather than a simulant. This table is descriptive and enters no multiplicity accounting.

| Agent | Series | $\log_{10} P$ | Nearest simulant | $\Delta$ | Nearest of pool | $\Delta$ |
|---|---|---|---|---|---|---|
| phosgene (CG) | choking | 3.152 | dichloromethane | 0.514 | dichloromethane | 0.514 |
| cyanogen chloride (CK) | blood | 3.090 | dichloromethane | 0.451 | dichloromethane | 0.451 |
| hydrogen cyanide (AC) | blood | 2.870 | dichloromethane | 0.232 | dichloromethane | 0.232 |
| chloropicrin (PS) | irritant | 1.380 | dichloromethane | 1.258 | water | 0.005 |
| sarin (GB) | G | 0.456 | DMMP | 0.473 | p-dichlorobenzene | 0.216 |
| lewisite 1 (L) | L | -0.237 | DPGME | 0.161 | DMSO | 0.015 |
| soman (GD) | G | -0.387 | DPGME | 0.011 | DPGME | 0.011 |
| sulfur mustard (HD) | H | -0.959 | DPGME | 0.561 | DPGME | 0.561 |
| tabun (GA) | G | -1.155 | DPGME | 0.757 | DPGME | 0.757 |
| cyclosarin (GF) | G | -1.357 | DPGME | 0.959 | DPGME | 0.959 |
| VX | V | -3.057 | DPGME | 2.659 | DPGME | 2.659 |

$\Delta$ is the gap in $\log_{10}(P/\mathrm{mmHg})$. The pool is the seventeen non-agents, so the last column is a simulant where one is nearest and an interferent otherwise.

Table S8: Equal weight per descriptor against equal weight per block, in the merged space. Valid-match counts are identical under the two weightings on every panel construction. The smallest $p$ is taken over the eighteen-test family of three spaces by six statistics, and $q$ is the Benjamini–Hochberg adjusted value of that family.

| | Valid matches | | Smallest nominal $p$ | | Smallest $q$ over 18 | |
|---|---|---|---|---|---|---|
| Configuration | per desc. | per block | per desc. | per block | per desc. | per block |
| Thirty-five, incumbent three | 0/11 | 0/11 | 0.192 | 0.061 | 0.630 | 0.630 |
| Thirty-five, incumbent 3 + literature 5 | 5/11 | 5/11 | 0.039 | 0.022 | 0.212 | 0.222 |
| Twenty-eight, incumbent three | 3/11 | 3/11 | 0.025 | 0.009 | 0.163 | 0.146 |

Table S9: Valid matches and the lewisite row across panel configurations (Section S21). A match requires a normalised distance of at most two *and* that no non-simulant lies closer.

| Configuration | Valid matches | Nearest simulant to lewisite | Normalised distance |
|---|---|---|---|
| Twenty-eight, incumbent three | 3/11 | dichloromethane | 2.14 |
| Thirty-five, incumbent three | 0/11 | dichloromethane | 2.35 |
| Thirty-five, incumbent 3 + literature 5 | 5/11 | CEES | 2.34 |
| Thirty-six, incumbent three | 0/11 | dichloromethane | 2.04 |
| Thirty-six, incumbent 3 + literature 6 | 6/11 | $AsCl_3$ | 1.25 |

Table S10: GFN2-xTB local reactivity at phosphorus for the six phosphorus-bearing compounds. $f^+$(P) is the condensed Fukui index for nucleophilic attack at phosphorus, in the frontier-electron formulation of Parr and Yang[S33] with the atomic condensation of Yang and Mortier.[S34] $\omega$ is the global electrophilicity index in the definition of Parr et al.[S35] with $\eta$ equal to half the HOMO–LUMO gap, and $\omega_k = \omega f^+$(P) is the composite local electrophilicity of Domingo et al.[S32] The two reactivity conventions disagree about VX and the text says why.

| Compound | $q_P$ (e) | $q_{P=O}$ (e) | $W_{P=O}$ | $f^+$(P) | $\omega$ (eV) | $\omega_k$ (eV) |
|---|---|---|---|---|---|---|
| GB | 0.54044 | -0.42847 | 1.9527 | 0.66793 | 5.69 | 3.80 |
| GD | 0.54460 | -0.43006 | 1.9507 | 0.63692 | 5.54 | 3.53 |
| GF | 0.53868 | -0.42662 | 1.9546 | 0.63203 | 5.62 | 3.55 |
| **DMMP** | 0.46630 | -0.44588 | 1.9246 | 0.55852 | 4.66 | 2.60 |
| GA | 0.34433 | -0.38982 | 2.0146 | 0.42143 | 20.73 | 8.74 |
| VX | 0.33196 | -0.40303 | 1.9863 | 0.20431 | 17.99 | 3.68 |

GB/GD/GF mean $f^+$(P) = 0.6456 ± 0.0195, DMMP 4.47 SD below; mean $\omega_k$ = 3.63 ± 0.15, DMMP 6.87 SD below. $\omega$ and $\omega_k$ are reported here and used in no distance (Section ). DMMP/VX = 2.73 by $f^+$(P) but 0.71 by $\omega_k$. Values are GFN2-derived and relative within one protocol.

Table S11: Premise test under the label-permutation null. The statistic is the mean over compounds of (agreement with itself across concentrations) minus (agreement with the other compounds' consensus patterns). The null permutes compound labels within each concentration. Fixing the first concentration's labels leaves $(n!)^{C-1}$ distinct relabellings, which is enumerable for the four-compound set and sampled for the others.

| Set | Channel | Observed | Null mean | Null s.d. | $p$ | Null built from |
|---|---|---|---|---|---|---|
| First | ultraviolet | +0.7288 | +0.1571 | 0.1013 | 0.0001 | 10,000 sampled |
| First | reflectance | +0.6931 | +0.1497 | 0.0845 | 0.0001 | 10,000 sampled |
| Second | ultraviolet | +0.5578 | +0.1024 | 0.0903 | 0.0001 | 10,000 sampled |
| Second | reflectance | +0.6187 | +0.1109 | 0.0654 | 0.0001 | 10,000 sampled |
| Third | ultraviolet | +0.5669 | +0.1898 | 0.1442 | 0.0046 | 13,824 enumerated |
| Third | reflectance | +0.6753 | +0.3138 | 0.1025 | 0.0033 | 13,824 enumerated |

Adjusted values are Benjamini–Hochberg over the six cells; the smallest is 0.0001 and all six clear $q = 0.05$.

Table S12: Regulatory acute inhalation values held for the three incumbent simulants, by register. Sources and collection dates are in `sources/registers_20261106.csv` of the deposit. EPA AEGL is the axis the agents are placed on in the main text; the other registers use different averaging times and severity definitions and are not interchangeable with it.

| Compound | Register | Tier | Value | Unit | Basis |
|---|---|---|---|---|---|
| DMMP | DOE PAC | PAC-1 | 11 | mg $m^{-3}$ | TEEL |
| DMMP | DOE PAC | PAC-2 | 120 | mg $m^{-3}$ | TEEL |
| DMMP | DOE PAC | PAC-3 | 730 | mg $m^{-3}$ | TEEL |
| DMMP | EPA AEGL | AEGL-3 10min | not listed | | – |
| DPGME | EPA AEGL | AEGL-3 10min | not listed | | – |
| DPGME | NIOSH IDLH | IDLH | 600 | ppm | acute toxicity data in humans and ar |
| DPGME | NIOSH REL | STEL | 150 | ppm | – |
| DPGME | NIOSH REL | TWA | 100 | ppm | – |
| DPGME | OSHA PEL | TWA | 100 | ppm | – |
| dichloromethane | DOE PAC | PAC-1 | 200 | ppm | AEGL |
| dichloromethane | DOE PAC | PAC-2 | 560 | ppm | AEGL |
| dichloromethane | DOE PAC | PAC-3 | 6900 | ppm | AEGL |
| dichloromethane | EPA AEGL | AEGL-3 10min | 41685 | mg $m^{-3}$ | – |

Table S13: Decoupling test. Fidelity is the standardised electronic-space distance to the nearest authentic agent; safety is the measured EPA AEGL-3 (10 min).

| Quantity | Value |
|---|---|
| Candidates with both axes measured | 207 |
| Spearman $r$ | +0.062 (95% CI −0.077 to +0.199) |
| Pearson $r$ | +0.072 (95% CI −0.092 to +0.225), $p = 0.30$ |
| Variance of toxicity explained by fidelity | 0.5% |
| Safety bar: dichloromethane, 41,685 mg $m^{-3}$ | $\log_{10} = 4.62$ |
| Faithful and safe at that bar | 0 |
| Faithful and safe one order of magnitude below it | 8 |
| clearing the structural-alert gate | 8 |
| more faithful than dichloromethane itself | 8 |
| Most faithful of them | Trichloroethylene (0.25 vs 0.78) |